TH series

Industrial Robot

INSTRUCTION MANUAL

INDUSTRIAL ROBOT

TRANSPORTATION AND INSTALLATION MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.5

April 1, 2020

Shibaura Machine

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Preface

This manual describes how to unpack and install the industrial robot and controller. Specifically, it describes how to unpack the shipment containing the equipment, how to install the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.

Before beginning the work according to this manual, read through the Startup Manual and Safety Manual so that you can learn the names and functions of various components and the safety measures.

This manual is divided into the following four (4) sections:

Section 1 Transportation
This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.

Section 2 Installation
This section discusses the equipment installation environment, space requirements, and how to install the equipment.

Section 3 System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 4 Tool Interlace
This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or major injuries".
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited(must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory(must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. The details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

DANGER	
Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, securely fix it on the base. The robot will be unstable if just set down, and it could tilt over.

Table of Contents

Page

Section 1 Transportation 8
1.1 Unpacking 8
1.2 Transportation 11
1.2.1 Mass and Dimensions 11
1.2.2 Transporting the Robot 11
1.2.3 Transporting the Controller 15
1.3 Storage 15
1.3.1 Storage Precautions for the Robot 15
1.3.2 Storage Precautions for the Controller 15

Section 2 Installation 16
2.1 Installation Environment 16
2.2 Robot Installation 17
2.2.1 External Dimensions 17
2.2.2 Working Envelope 17
2.2.3 Coordinate System 21
2.2.4 Installing the Robot 25
2.3 Installing the Controller 28
2.3.1 External Dimensions 28
2.3.2 Rack Mounting Dimensions 29
2.3.3 Precautions for Rack Mounting 30
2.3.4 Precautions for Direct Installation 32
2.4 Precautions for Handling the Teach Pendant 33
2.5 Safety Measures 34
2.6 Position Detector Backup Batteries 35
2.7 Using the Robot in Clean Room 37

Section 3 System Connections 38
3.1 Cable Wiring 38
3.1.1 Connector Arrangement on the Controller 38
3.1.2 Connecting the Power Cable 39
3.1.3 Connecting the Motor Cable 41
3.1.4 Connecting the Encoder Cable 43

Page

3.1.5 Connecting the Hand Control Signal Cable 43
3.1.6 Grounding the Robot 43
3.1.7 Inserting and Removing Cables 44
3.1.8 Examples of Connector Terminal Arrangement 46
3.2 Controller Connector Signals 48
3.2.1 Connector Signal Connection Diagrams 48
3.2.2 Jumpers for Safety Related Signals 48

Section 4 Tool Interface 50
4.1 Mounting Tool 50
4.2 Tool (Hand) Wiring and Piping 52
4.2.1 Controller's Tool Signals 55
4.3 Permissible Load Conditions and Program Setting 58
4.3.1 Permissible Load Conditions 58
4.3.2 Load Conditions and Program Setting 60
4.3.3 Setting Robot Acceleration and Maximum Speed for Load Conditions 64

Section 5 Specifications 73
5.1 Specifications Table 73

Appendix A Calculating Control Power Capacity 77

Section 1

Transportation

1.1 Unpacking

The robot and controller are shipped separately in corrugated cardboards. The teach pendant and other accessories such as the instruction manual are packaged separately.

Open the packages in a location easily accessible to where the equipment is to be installed. Take careful precautions not to damage the robot or controller. After opening the packages, make sure that all the accessories are present and that nothing has been damaged in transit.



DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the corrugated cardboards, polyethylene shipping bags and cushion material according to your standards.

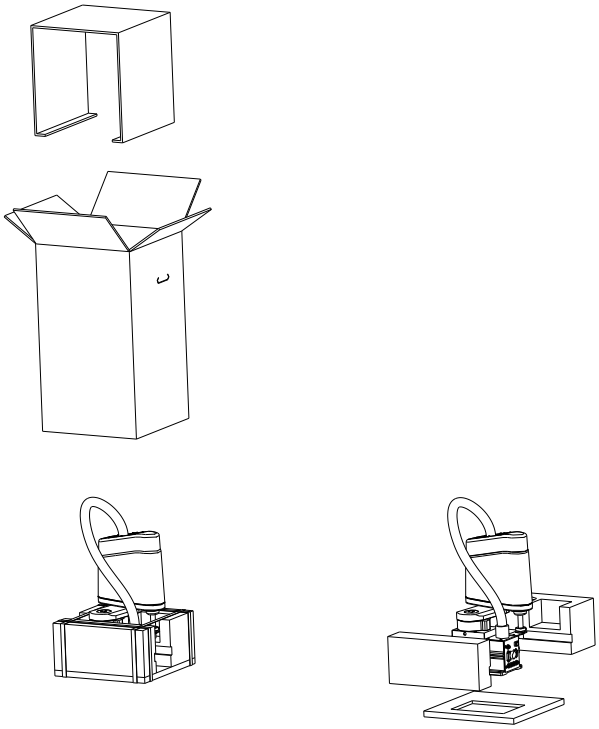


Fig. 1.1 Robot package (TH250A)

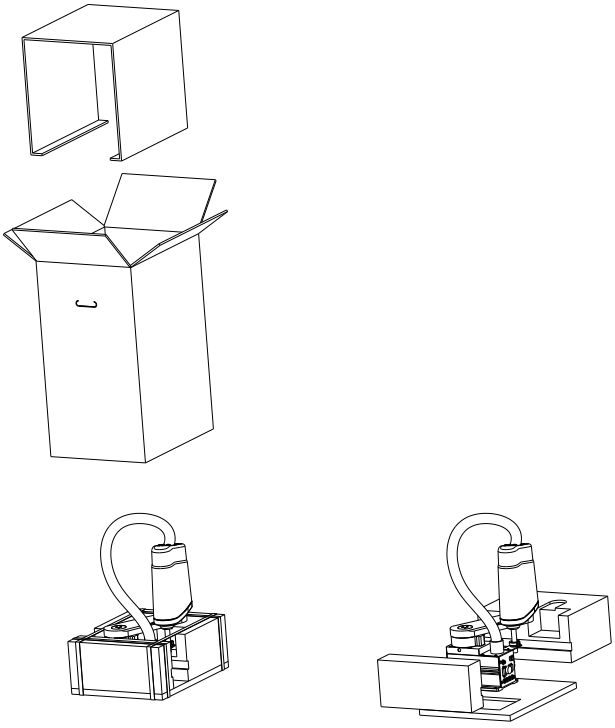


Fig. 1.2 Robot package (TH350A)

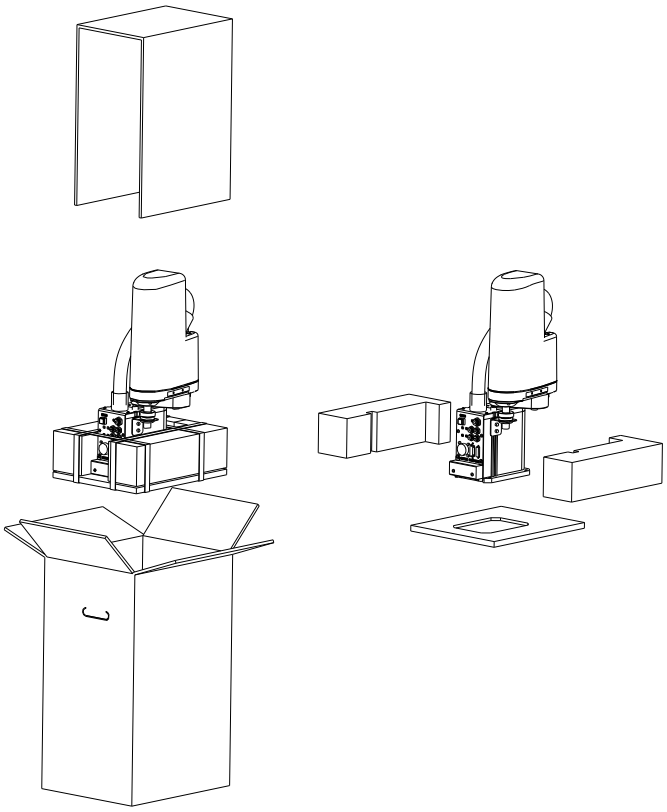


Fig. 1.3 Robot package (TH180)

1.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact ($9.8 m/s^2$ or more) or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the nuts and screws completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

1.2.1 Mass and Dimensions

The mass and dimensions of the robot and controller are shown in Table 1.1.

Table 1.1

Type	Mass
Robot TH180	9 kg
Robot TH250A	14 kg
Robot TH350A	14 kg
Robot TH350A‐T	14 kg
Controller TS1000	10 kg

1.2.2 Transporting the Robot

To transport the robot, fold back and secure the arm with the attached clamp. (The robot is shipped in this posture. After you have unpacked the shipment, you can move it as it is.)

! DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

! CAUTION

- When transporting or unpacking the robot and robot controller, take careful precautions not to get injured by the drop of them or not to damage the equipment.
- When lifting the robot by workers, hold the locations (shaded areas) by hands as shown in Figs. 1.4 to 1.6. If the arm 2 or cover is held by hands, an unusually large force is exerted, resulting in a malfunction.
- If the robot has been carried manually by workers, take utmost care to prevent their hand or leg from being caught in the robot when placing it on the floor.

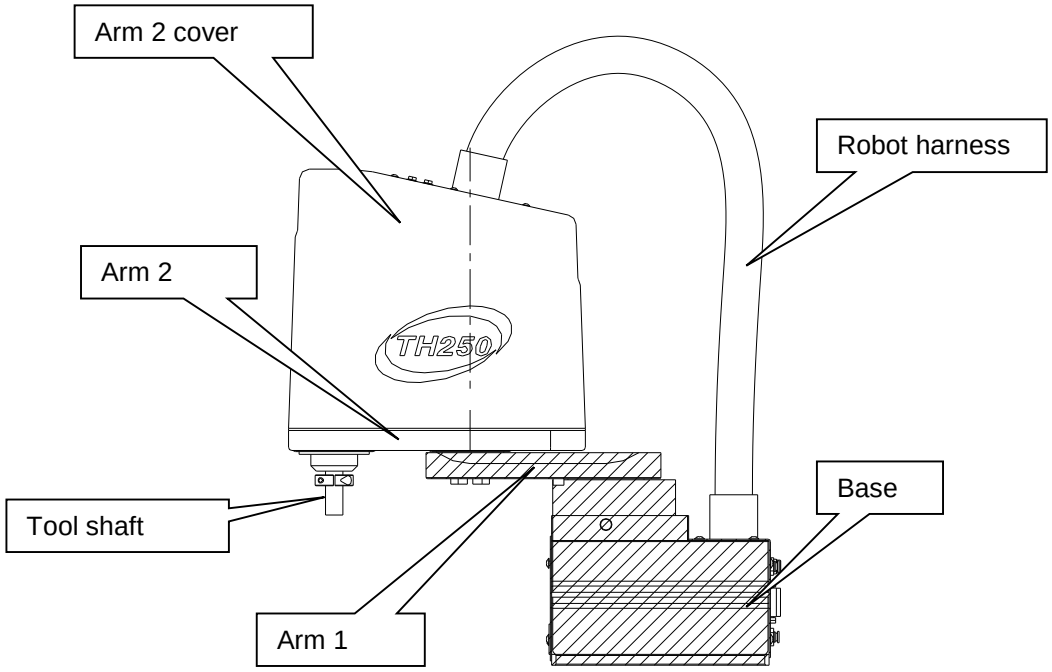


Fig. 1.4 Robot handling area (shaded area) (TH250A, TH350A)

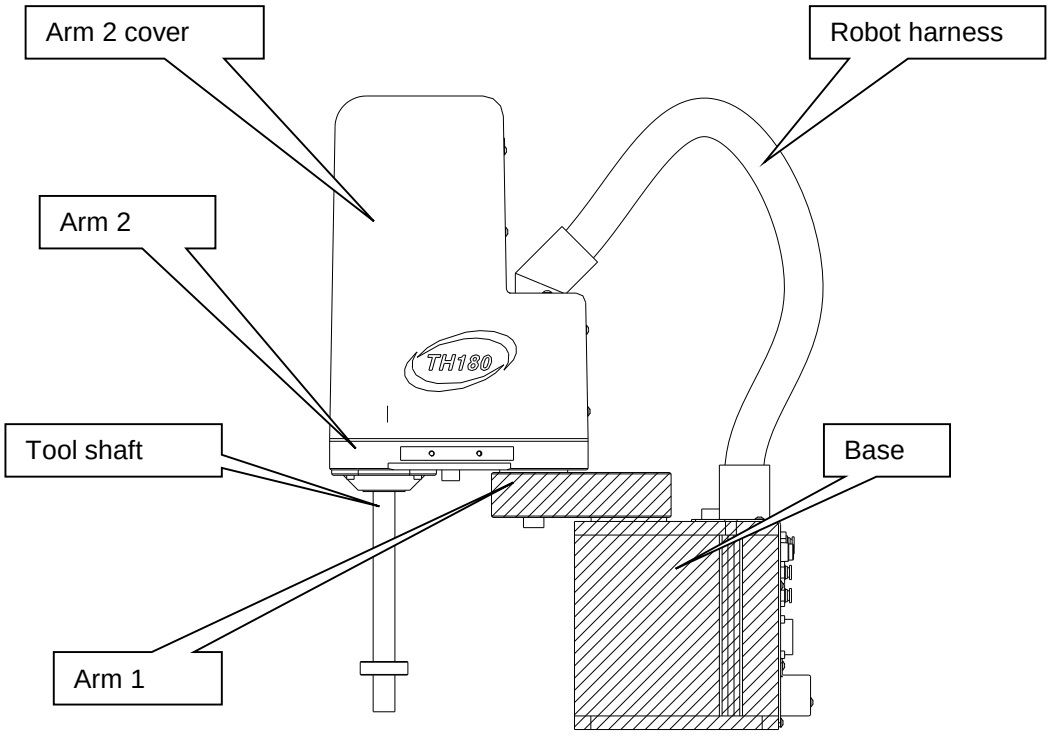


Fig. 1.5 Robot handling area (shaded area) (TH180)

After the installation, remove the clamp used for transport.

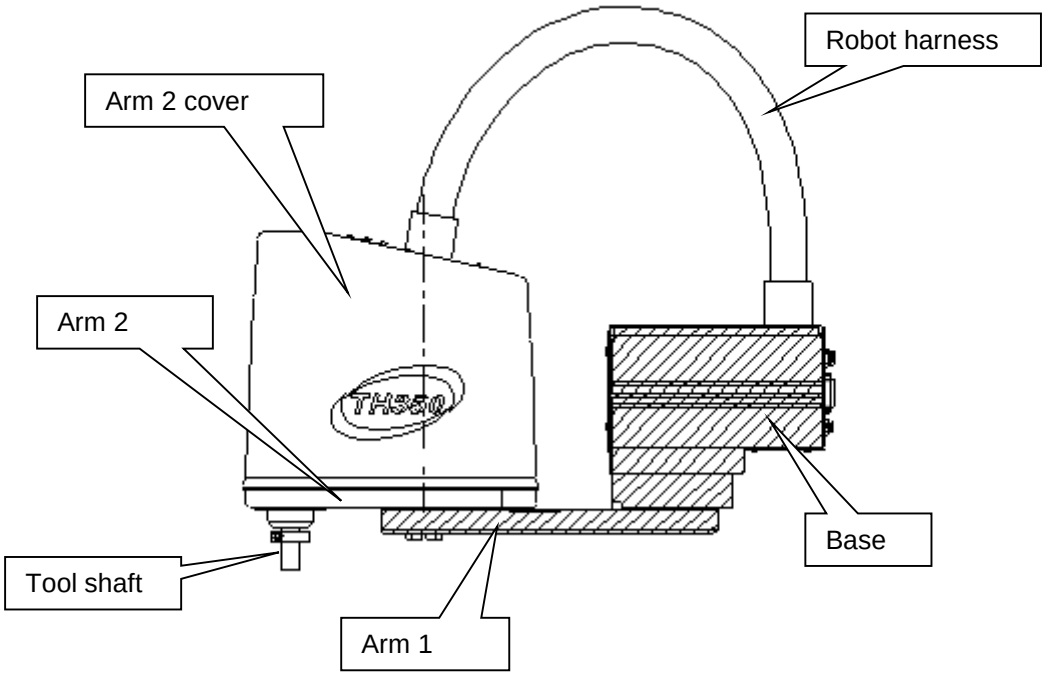


Fig. 1.6 Robot handling area (shaded area) (TH350A‐T)

1.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to catch your hands or feet.

1.3 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

1.3.1 Storage Precautions for the Robot

It is recommended to secure the robot in the shipment posture, using the attached clamp for transport.

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Keep the robot out of direct sunlight. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease.

1.3.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.

Section 2

Installation

2.1 Installation Environment

Table 2.1 shows the environmental conditions for the location in which the TH series Robot and TS1000 controller are to be installed.

Table 2.1 Environmental conditions for robot and controller

Item	Specifications
Temperature	In operation : 0 to 40°C In storage : –10 to 50°C
Humidity	20 to 90 % (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation : 0.98m/s^2 or less In transport : 9.8m/ s^2 or less
Dust	No inductive dust should exist. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment.
Gas	No corrosive or combustible gas should exist.
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.



DANGER

- Do not place the robot or controller near combustible. Doing so could lead to fires if it ignites due to a fault, etc.

⚠ CAUTION

- In the case where batteries for detecting the motor position are of alkaline type, the batteries can overheat, leak battery fluid, or rupture when used under high temperatures. Also, high temperatures can reduce the performance and lifespan of the battery. If using the robot under high temperatures, please consult with the SHIBAURA MACHINE sales office.

2.2 Robot Installation

Before installing the robot, you should plan a layout, fully considering the working envelope, coordinate system and space for maintenance.
It is recommended to replace the batteries at the time of installation so that they can be replaced at regular inspection and maintenance. For the battery replacement, see Para. 2.3.8 of the Maintenance Manual.

2.2.1 External Dimensions

External view drawings of the robot are shown in Figs. 2.1 to 2.4.

2.2.2 Working Envelope

Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by mis-operation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits which can be set by the user are provided.

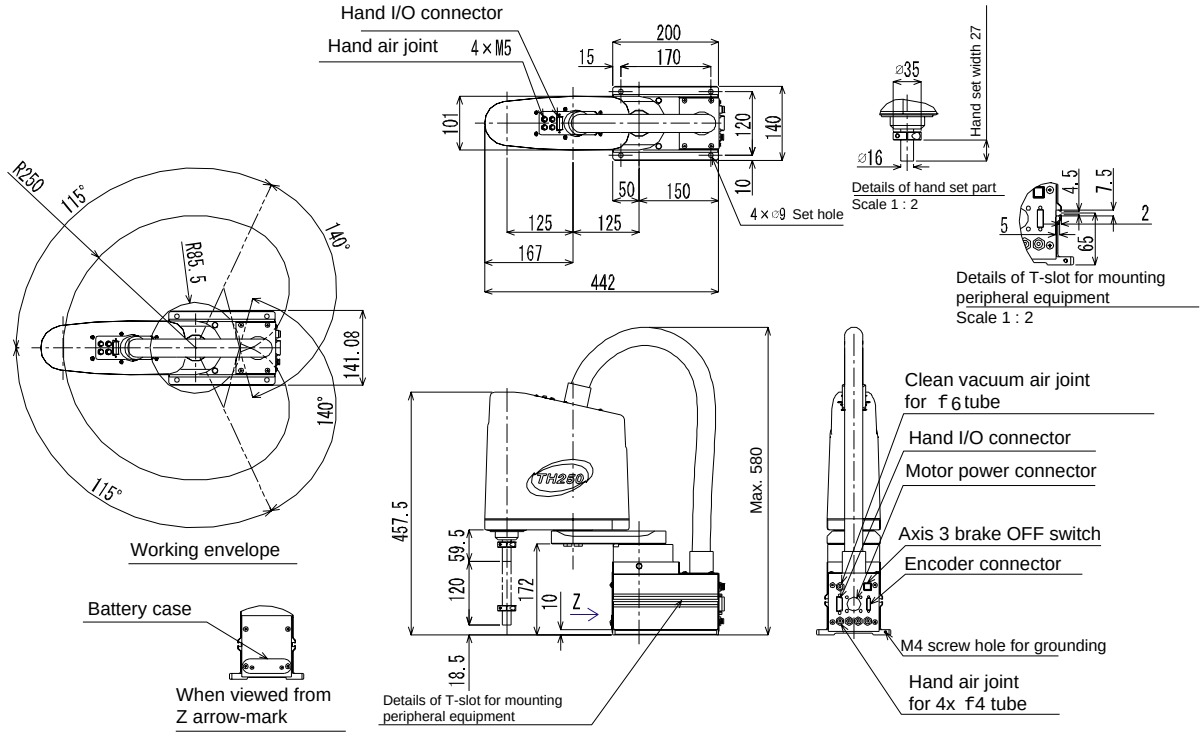


Fig. 2.1 External view of TH250A robot

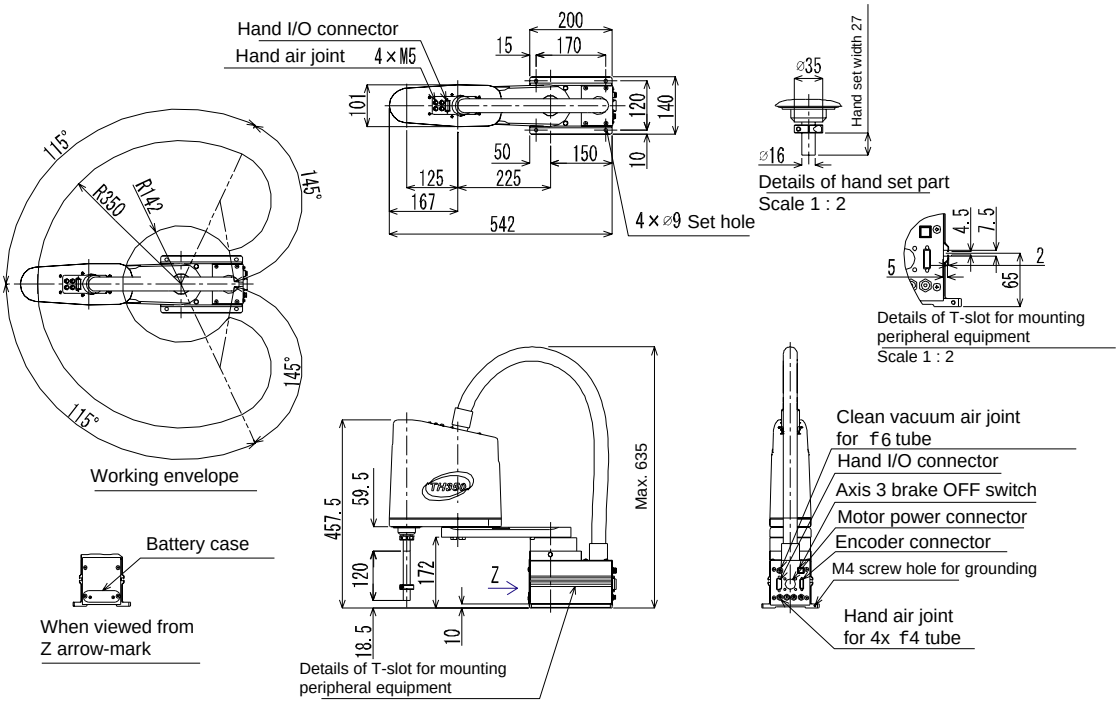


Fig. 2.2 External view of TH350A robot

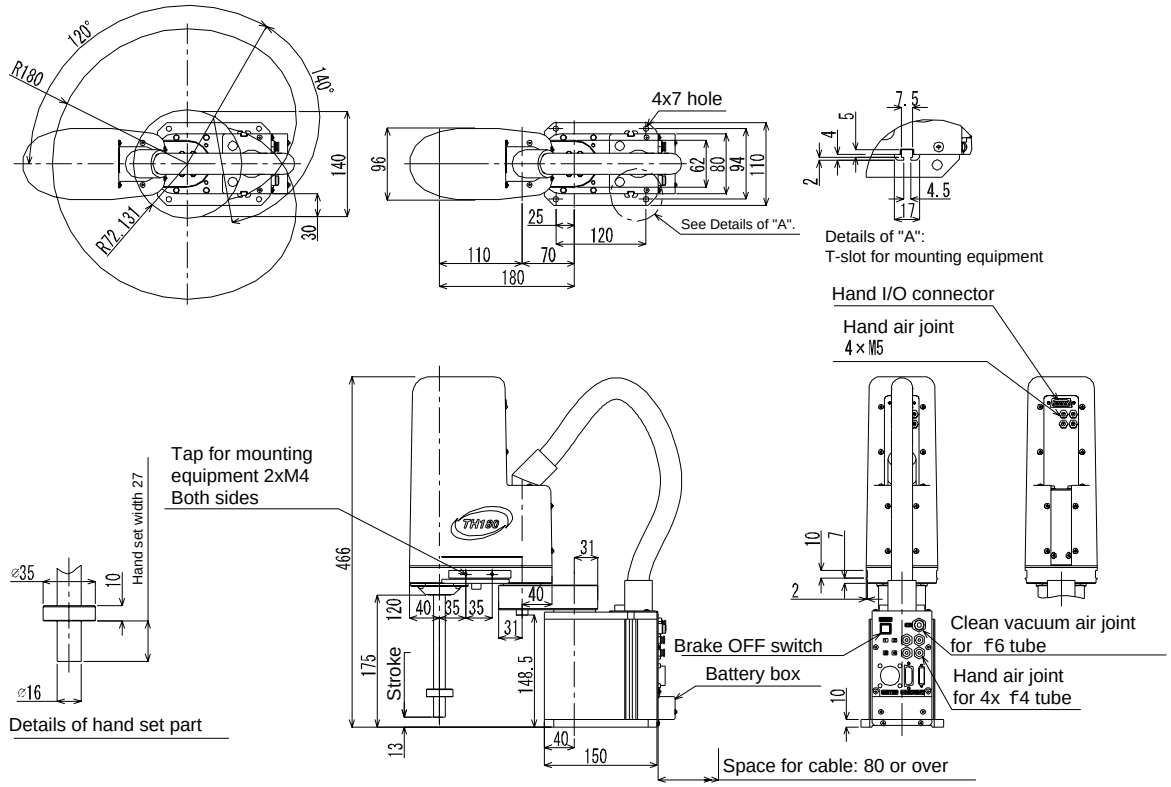


Fig. 2.3 External view of TH180 robot

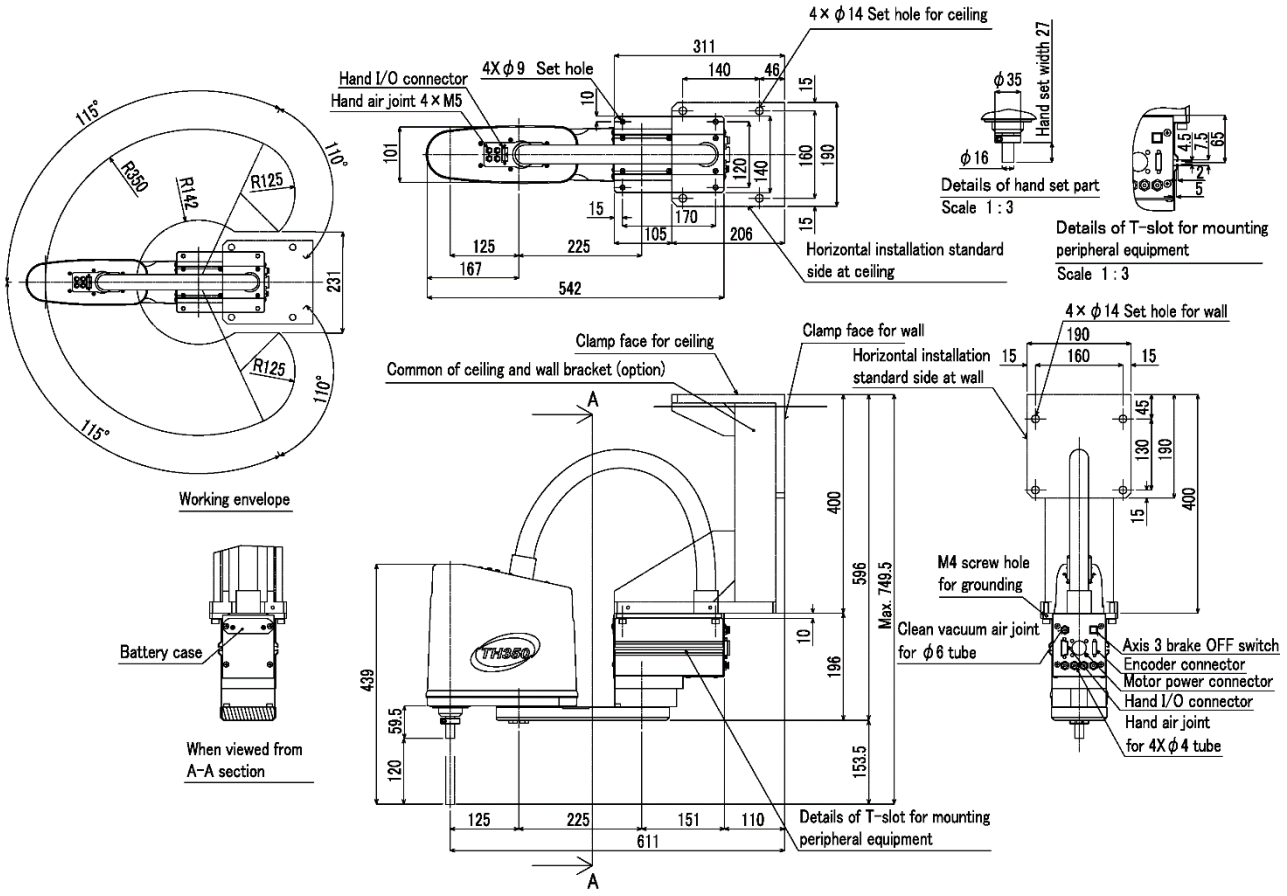


Fig. 2.4 External view of TH350A‐T robot

2.2.3 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. The base coordinate system is determined according to this calibration. Figs. 2.5 to 2.8 show the base coordinate system and zero position of each axis joint angle. Figs. 2.5 to 2.8 show the base coordinate system by $X_BY_BZ_B$.

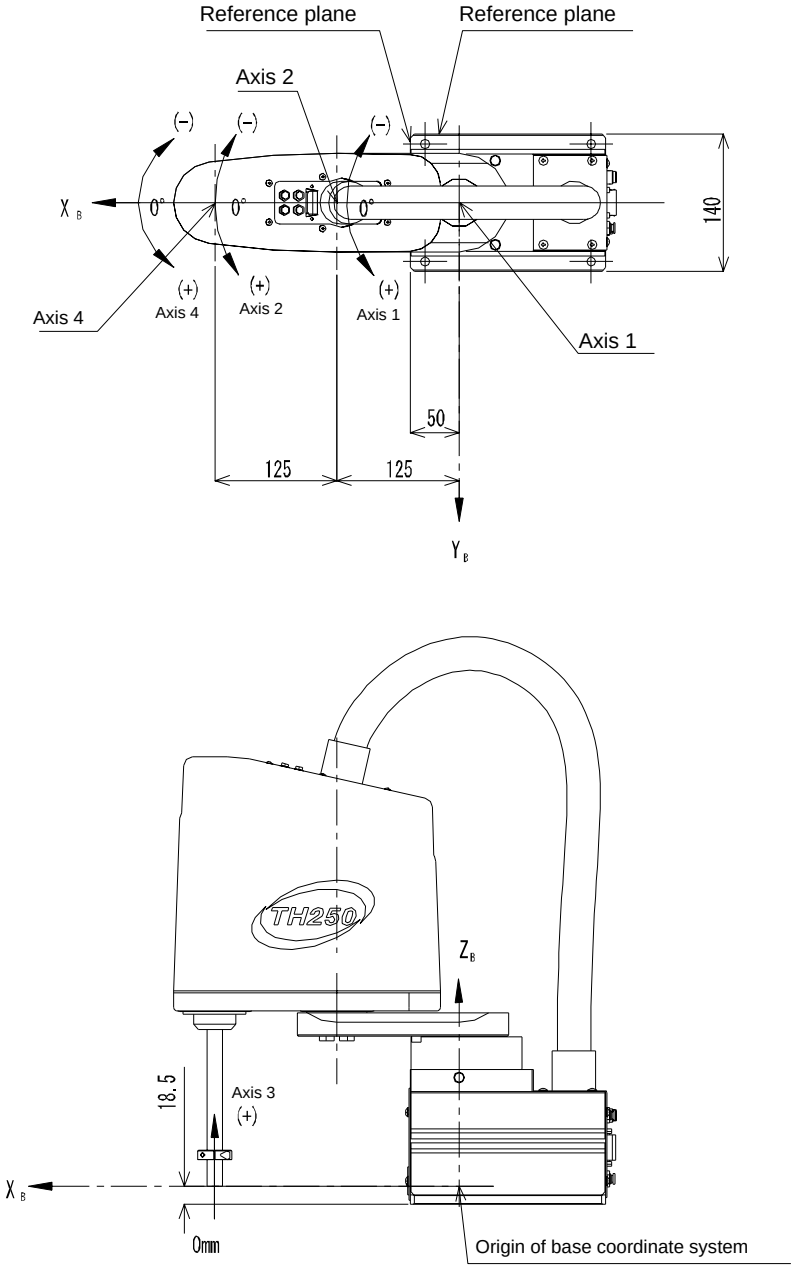


Fig. 2.5 TH250A base coordinate system and joint angle origin

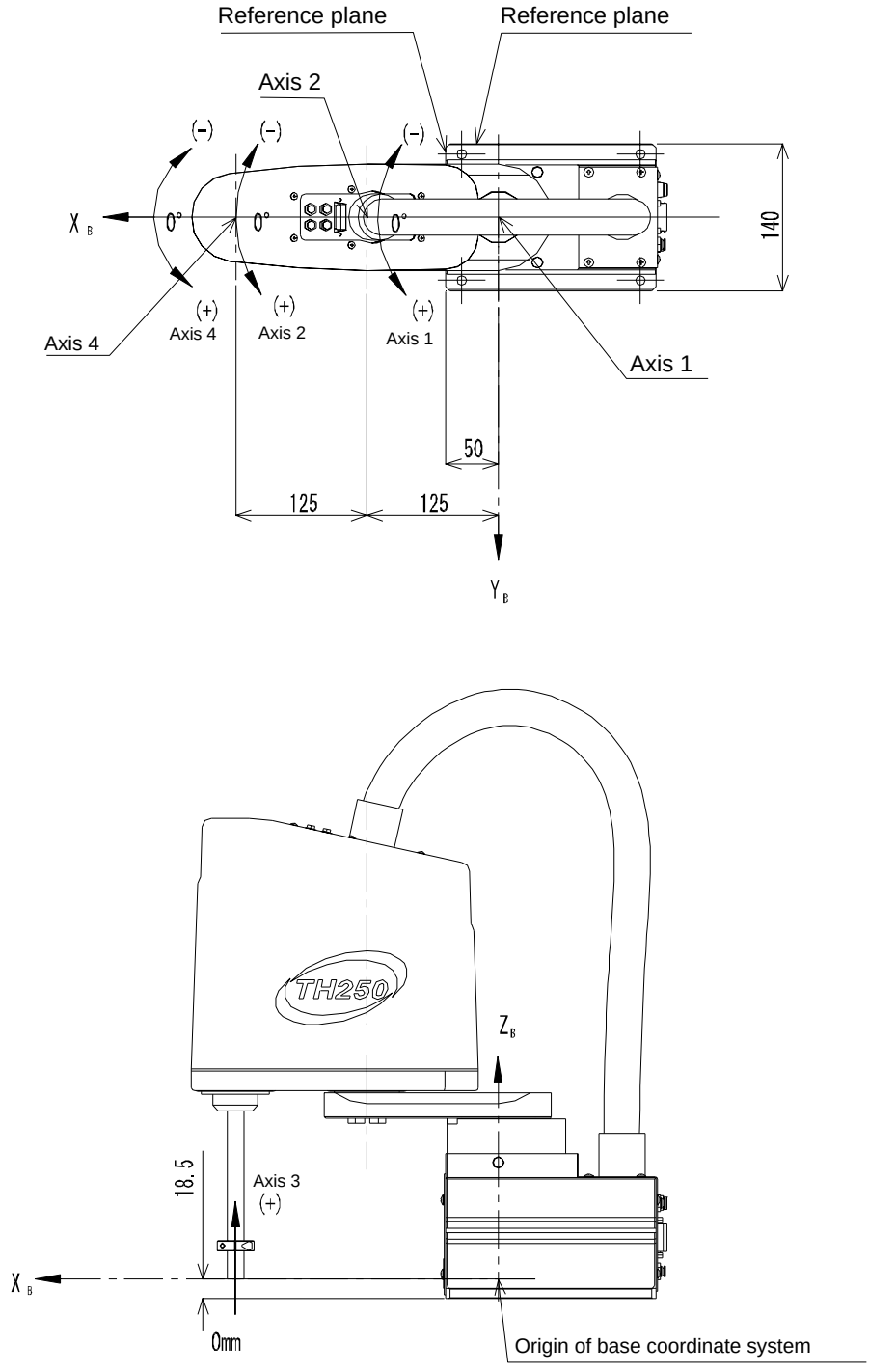


Fig. 2.6 TH350A base coordinate system and joint angle origin

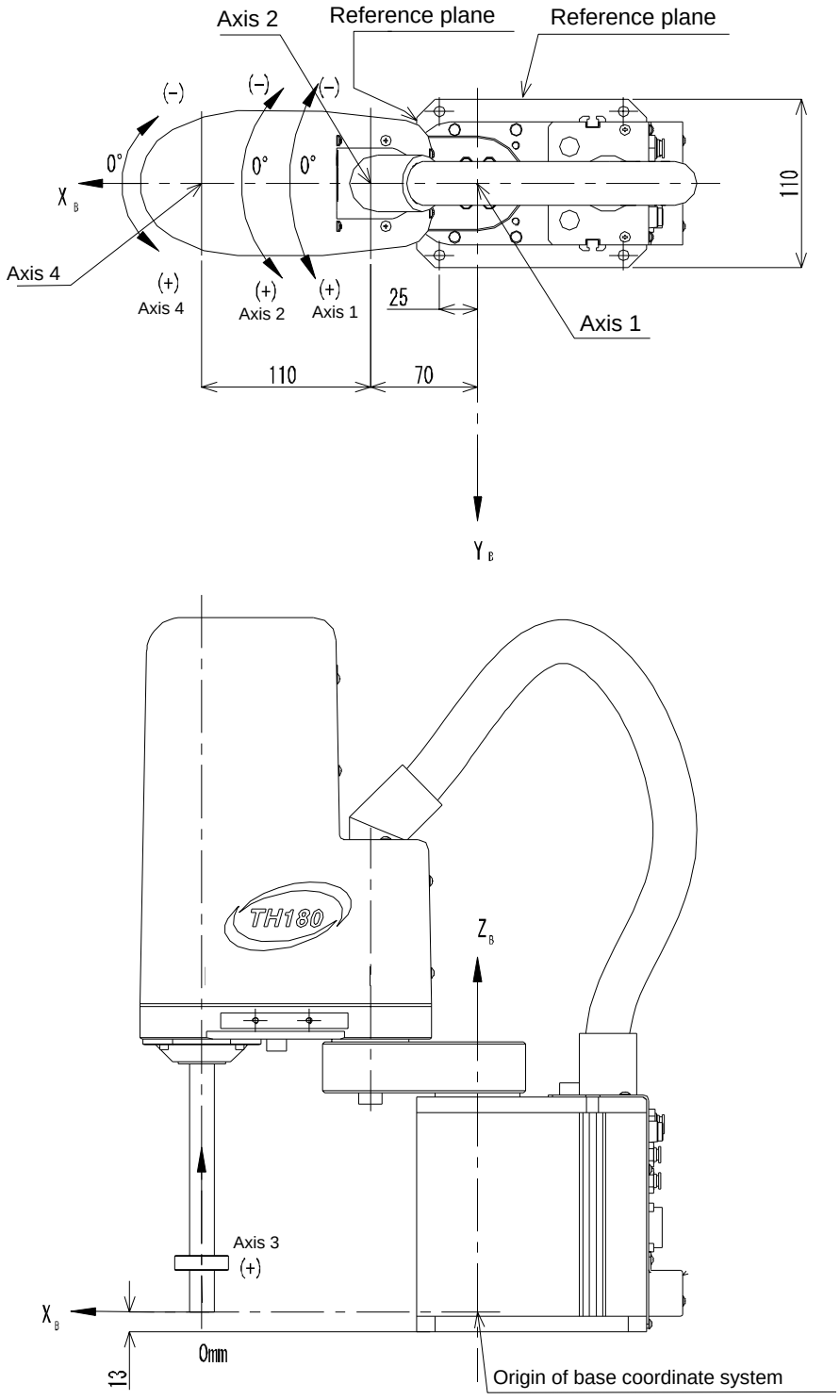


Fig. 2.7 TH180 base coordinate system and joint angle origin

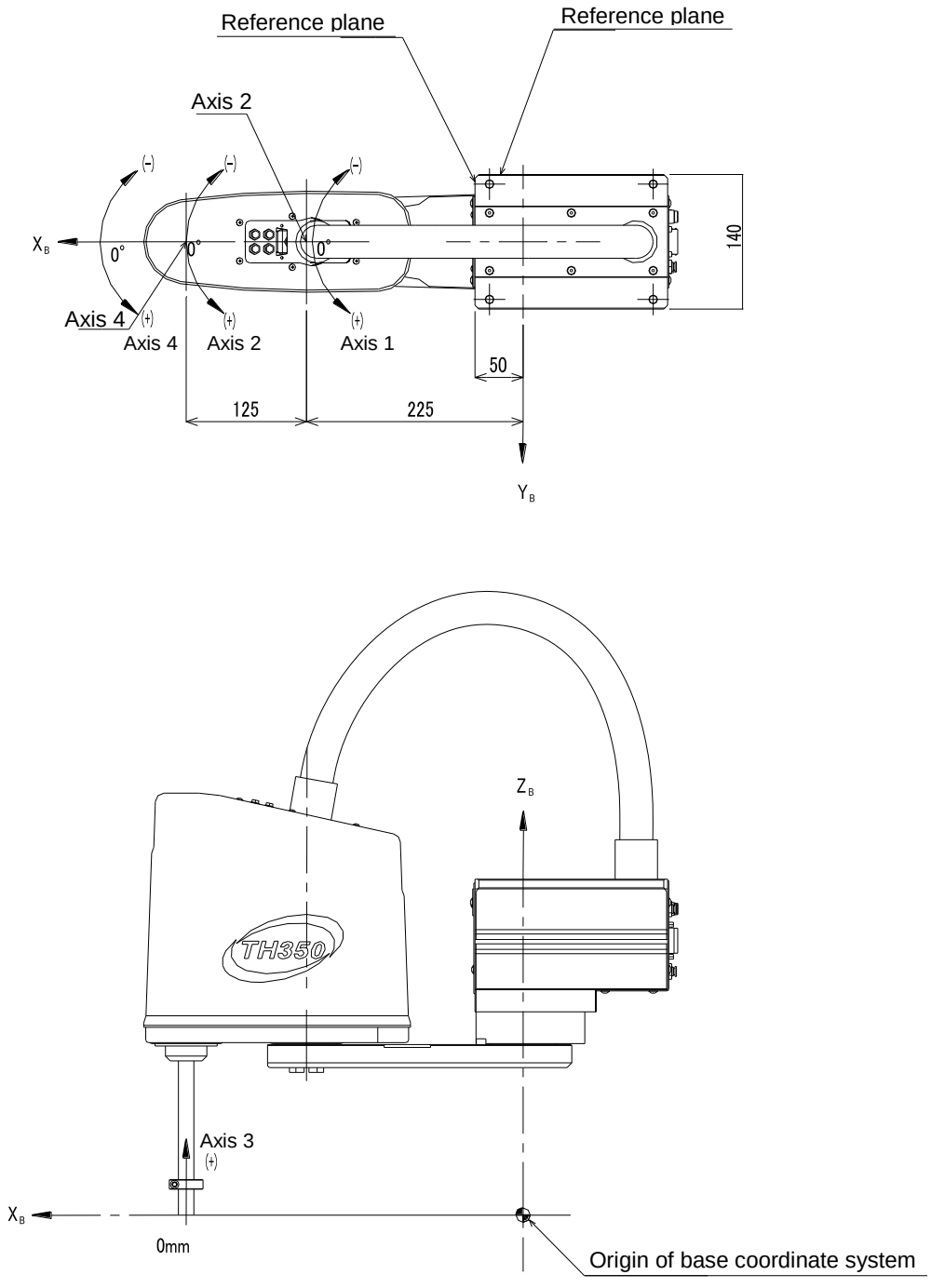


Fig. 2.8 TH350A‐T base coordinate system and joint angle origin

2.2.4 Installing the Robot

The robot is secured, using the set holes on the base. Use M8 hexagon socket head cap screws.
The robot installation method is shown in Figs. 2.9 to 2. 11. Reference planes are provided on the base section and marked with “xxx”.1
To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

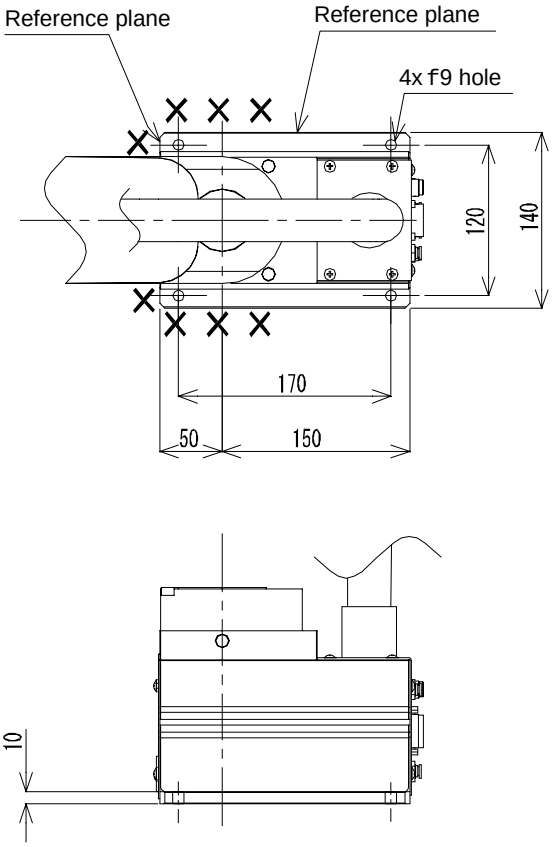


Fig. 2.9 Installing the TH250A and TH350A robots

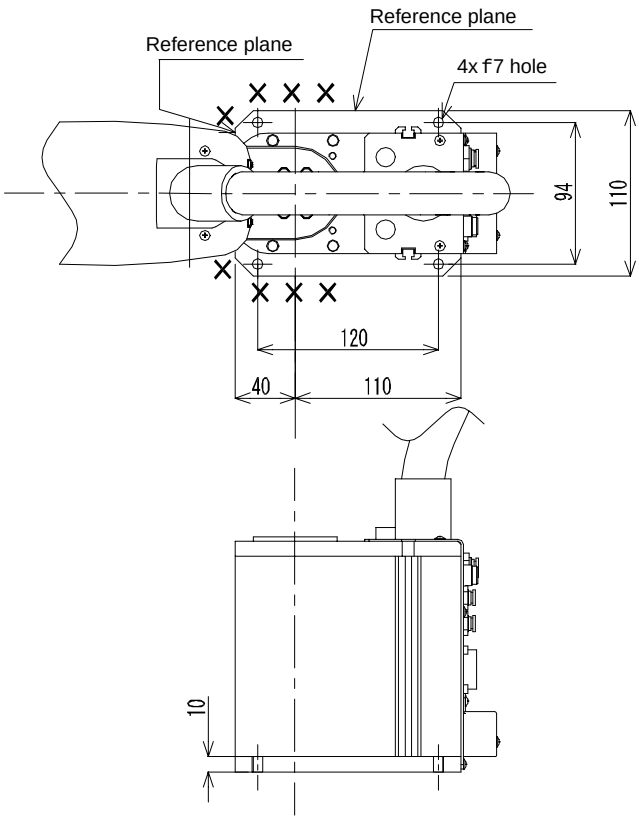


Fig. 2.10 Installing the TH180 robot

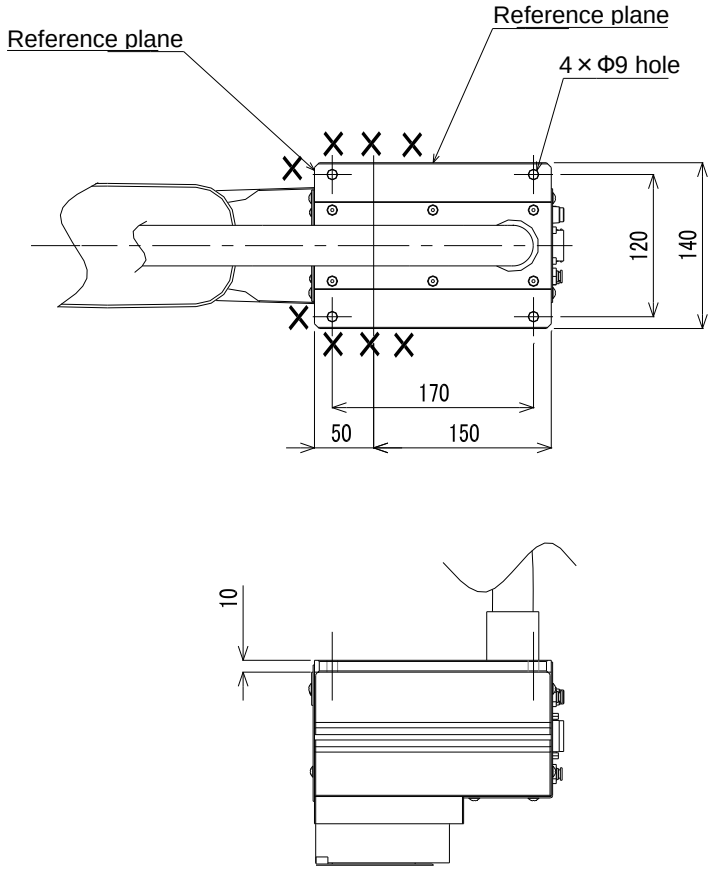
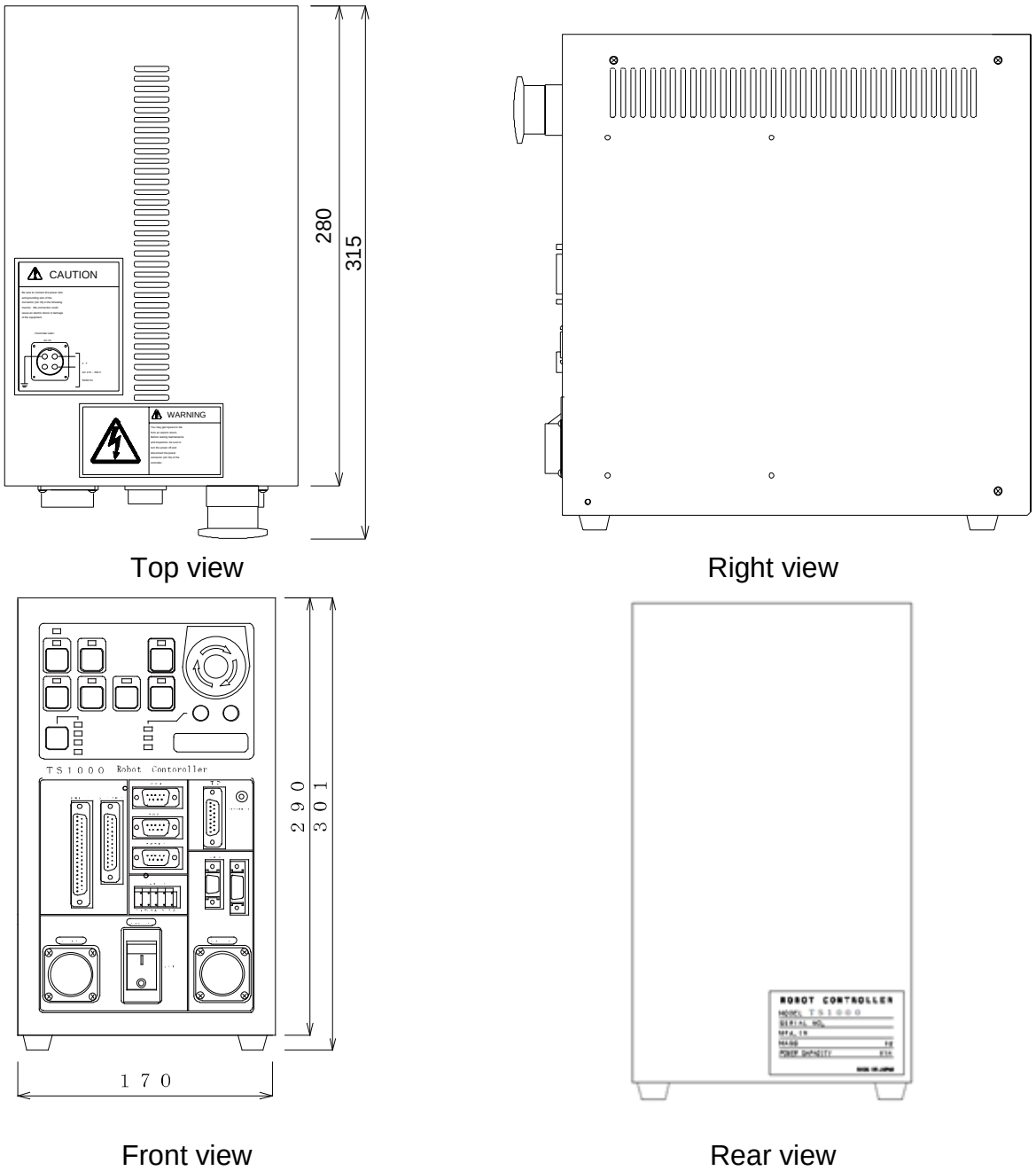


Fig. 2.11 Installing the TH350A‐T robot

2.3 Installing the Controller

2.3.1 External Dimensions

External view of the TS1000 controller is shown in Fig. 2.12.



Top view

Right view

Front view

Rear view

Fig. 2.12 External view of TS1000 controller

2.3.2 Rack Mounting Dimensions

When mounting the TS1000 robot controller in a rack, set the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets [1] in Fig. 2.13 are optional.

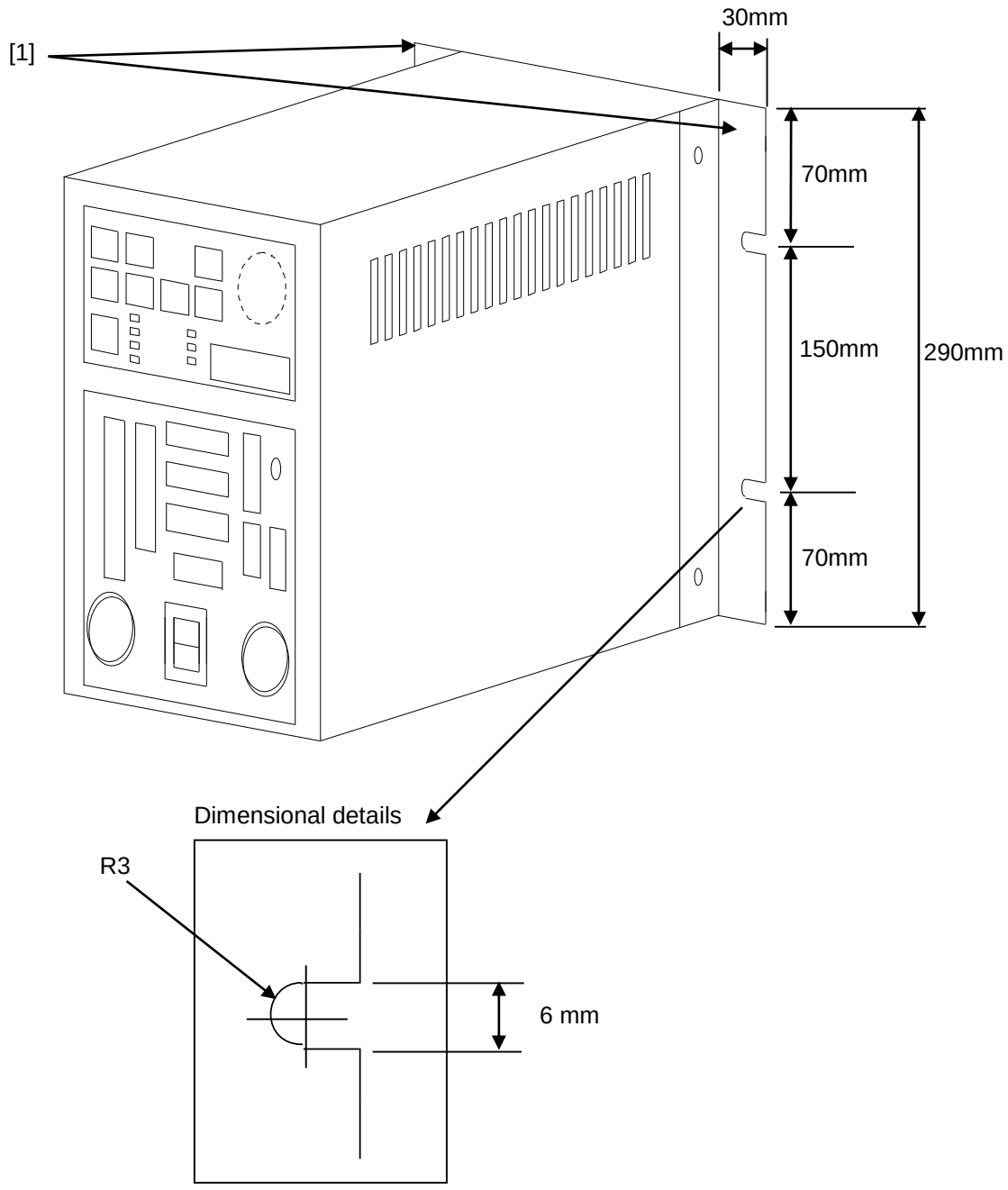


Fig. 2.13 Screw hole dimensions for securing controller

2.3.3 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the TS1000 robot controller.

(a) Use the screw holes provided on both ends of the front panel, and secure the controller. (Optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.
  If the heat is trapped in the rack or controller, faults could occur.

(b) As the cable connectors are connected to the front side of the controller, provide a space of 130 mm on the front side.
For maintenance, the upper cover should be removed. (See Fig. 2.14.)

Keep this in mind when installing the controller. At maintenance, the controller should be removed from the rack. Specifically, be careful of the following points.

1) Arrange the cables around the front side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

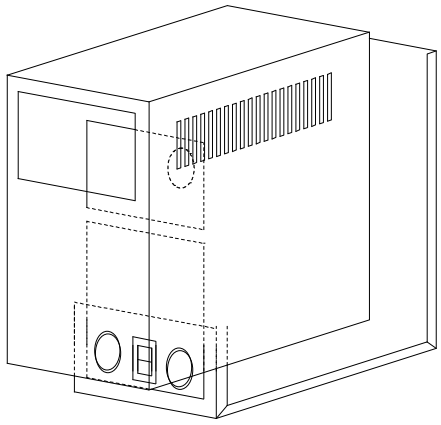


Fig. 2.14 Removing upper cover

(c) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with the screws.

(d) On the front of the controller, a clearance of approx. 130 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 50 mm is required for connecting a dummy plug.

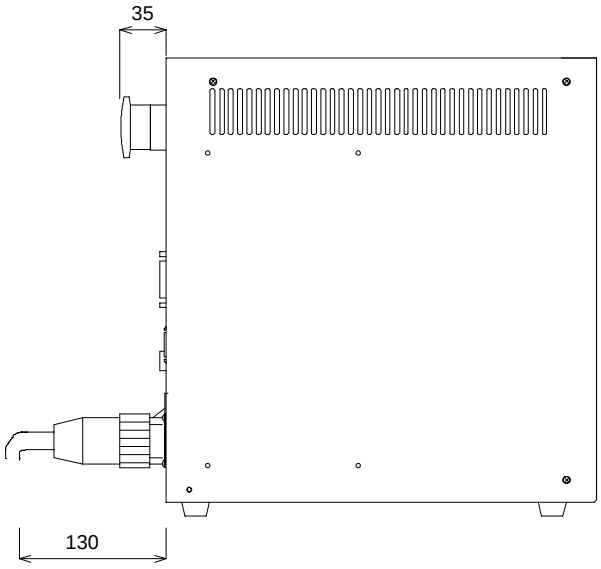


Fig. 2.15 Clearance of controller front side

2.3.4 Precautions for Direct Installation

When two (2) or more controllers are used, it is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction between the controllers.

! CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.
  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers. DO NOT place any object on top of them.

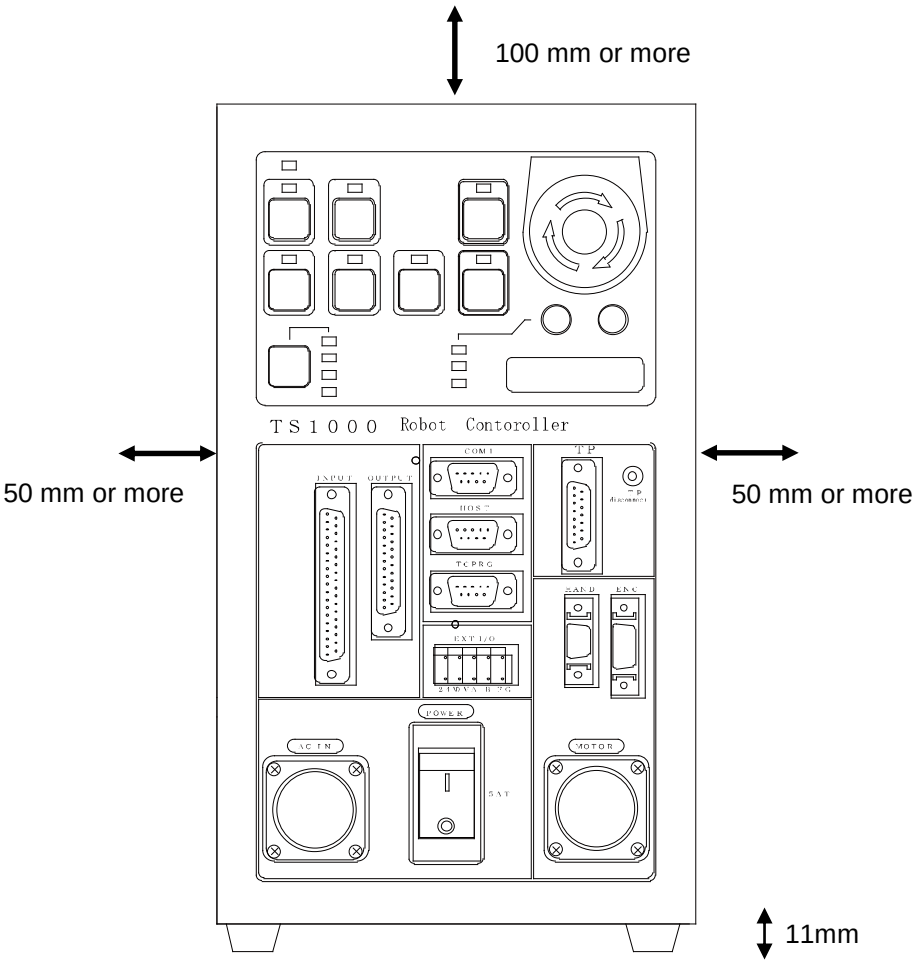


Fig. 2.16 Controller ventilation space

2.4 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

! CAUTION

- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

2.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide a limit switch, photo switch or foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent close contact (closed in normal operation) with compulsive opening function and must not be automatically recovered.

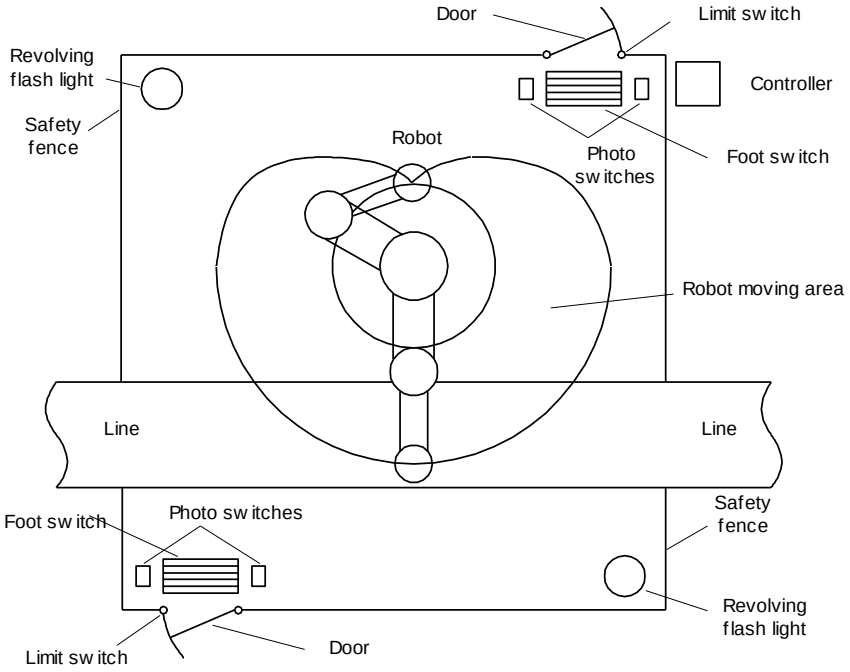


Fig. 2.17 Example of safety measures

d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

2.6 Position Detector Backup Batteries

The robot has batteries to back up positional information on position detectors in its base. Unless the robot is used for a long period of time, the battery voltage will drop and the positional data of origin will be lost.
Replace the batteries every year. It is recommended to replace the batteries at the time of installation so that they can be replaced at regular inspection and maintenance.
Also, if the battery voltage for backup of the X8LC printed board SRAM of controller has dropped, absolute position data set by the user will be lost. To avoid this, replace the batteries on a regular basis.
If the master power voltage of the batteries has dropped, an alarm of level 1 as shown below occurs first. When this happens, replace the batteries immediately. For the battery replacement, see Para. 2.3.8 and 3.3.3 of the Maintenance Manual.

Table 2.2 List of level 1 alarms

Error code	Descriptions
1–049	Axis1 Battery alarm
1–065	Axis2 Battery alarm
1–081	Axis3 Battery alarm
1–097	Axis4 Battery alarm
1–113	Axis5 Battery alarm
1–129	Axis6 Battery alarm
1–145	MAIN Battery alarm

If the master power voltage has dropped further, an alarm of level 8 as shown below occurs. When this happens, replace the batteries with new ones and perform encoder error reset operation because there is a fear that the position encoder backup data has been lost. For the encoder error reset operation, see Appendix B of the Maintenance Manual.

Table 2.3 List of level 8 alarms

Error code	Descriptions
8–065	Axis1 Encoder abnormal
8–097	Axis2 Encoder abnormal
8–129	Axis3 Encoder abnormal
8–161	Axis4 Encoder abnormal
8–193	Axis5 Encoder abnormal
8–225	Axis6 Encoder abnormal

2.7 Using the Robot in Clean Room

The main robot is designed pursuant to the Clean Class 10.
To be more specific, when this robot is operated in the clean bench (Clean Class 1 or less) installed in the downflow clean room, the number of 0.3 μm or more fine particles in 1 ft^3 as measured on the surface of the axis 3 stroke lower limit position is 10 or less.
The downflow air speed should be 0.3 to 0.5 m/s.
As the TS1000 controller is not designed to satisfy the specified clean class, take careful precautions to select the place of installation.
In the clean robot, an air tube is incorporated for ventilation. To operate it as the clean robot, scavenge by connecting a scavenging vacuum pump (or equivalent, which is to be provided by the customer) with the air joint. The target scavenging volume is 50 L/min.
When the robot is operated without scavenging, the specified clean class cannot be maintained.

Section 3

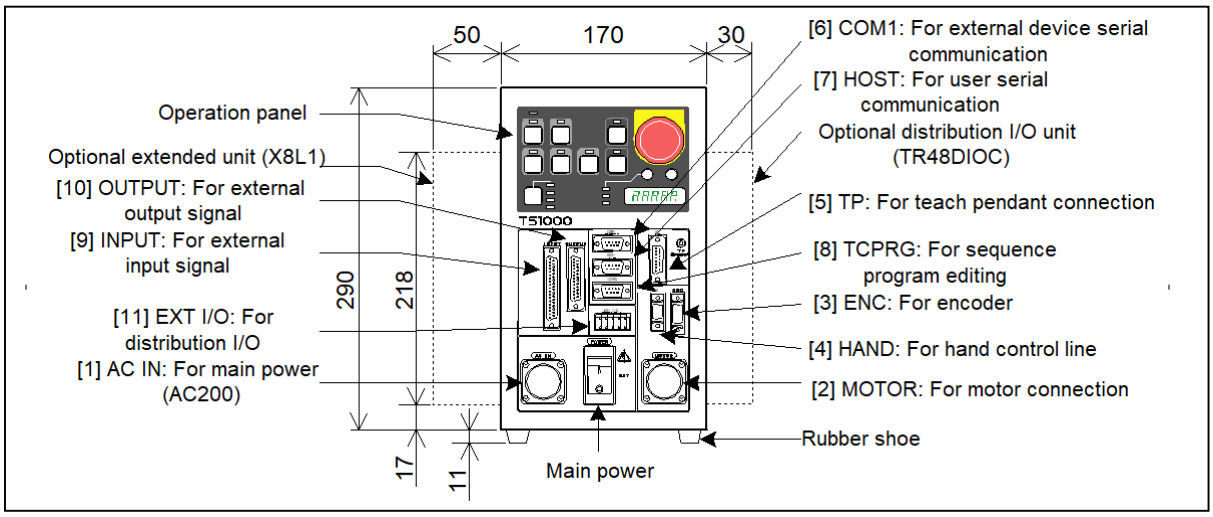
System Connections

3.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

3.1.1 Connector Arrangement on the Controller

The cables connected to the TS1000 robot controller are shown in Fig. 3.1.



Front view

Fig. 3.1 TS1000 robot controller connector layout

In the subsequent paragraphs, we explain how to connect cables [1] to [5]. For information on how to connect cables [6] to [11], refer to the Interface Manual.

3.1.2 Connecting the Power Cable “ACIN”

([1] of Fig. 3.1; plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 3.1 Power supply specifications

Item	Specifications
Power supply	Single phase, AC 190 ~ 250 V, 50/60 Hz±1 Hz
Power capacity	TH250A (0.9 kVA) TH350A (0.9 kVA) TH180 (0.5 kVA) TH350A–T (0.9 kVA)
Instantaneous power failure	Within half the cycle time
Grounding	Grounding with grounding resistance of 100 Ω or less

The connector is ACIN ([1] of Fig. 3.1).
The cable is not an accessory, so use the attached plug connector connected to ACIN on the controller side to manufacture a cable.

ACIN plug connector	Type: JL04V–6A18–10SE–EB	Maker:	Japan Aviation Electronics Industry
ACIN cable clamp	Type: JL04–18CK	Maker:	Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2 (AWG22 ~ 20)		

Wires are to be soldered to the connector (JL04V–6A18–10SE–EB).

! DANGER

- Always use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 3.1.7.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of “8–027 Slow Charge error” occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.8 of the Operator’s Manual.

3.1.3 Connecting the Motor Cable “MOTOR”

([2] of Fig. 3.1) (Cable attached)

The motor cable connects the controller and robot, supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot and turns on and off the brake for securing the motor axis. The connector for connecting the motor cable is MOTOR ([2] of Fig. 3.1). Location of the motor cable on the robot side is [1] in Fig. 3.2, [1] in Fig. 3.3 or [1] in Fig. 3.4.

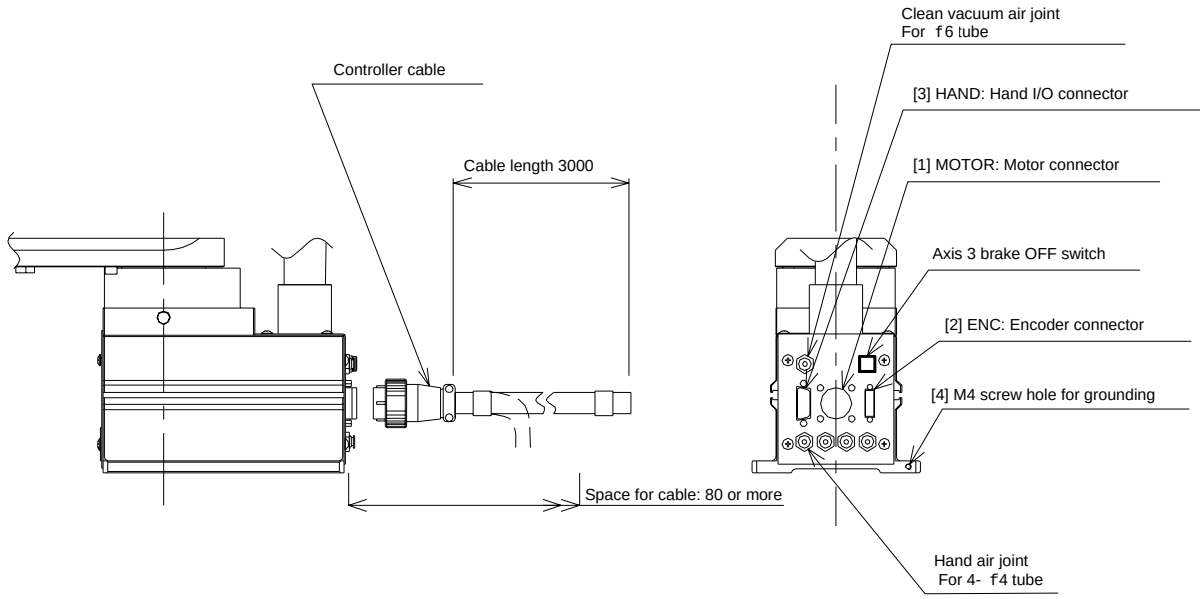


Fig. 3.2 Robot side connectors arrangement (TH250A, TH350A)

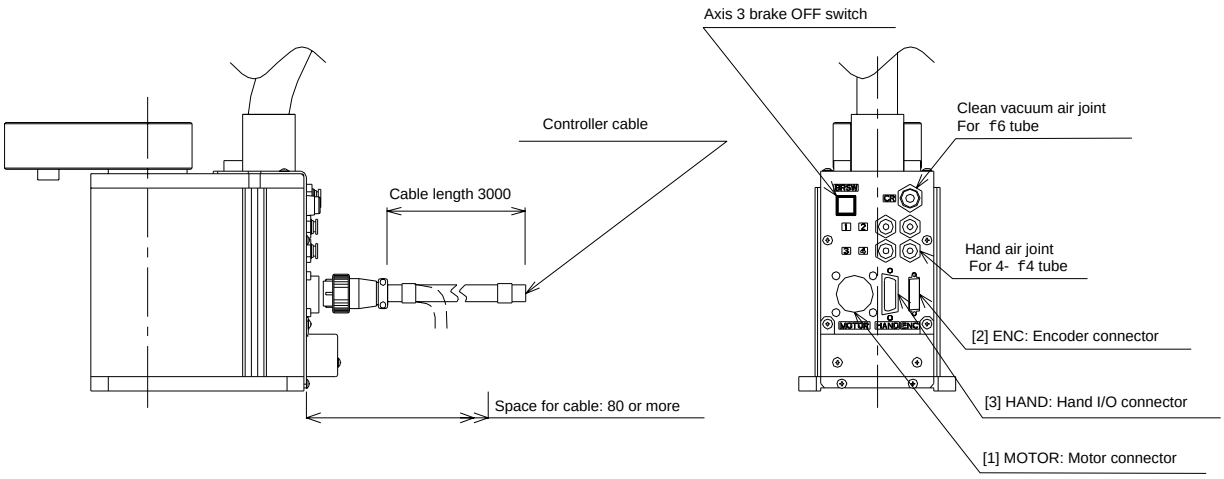


Fig. 3.3 Robot side connector arrangement (TH180)

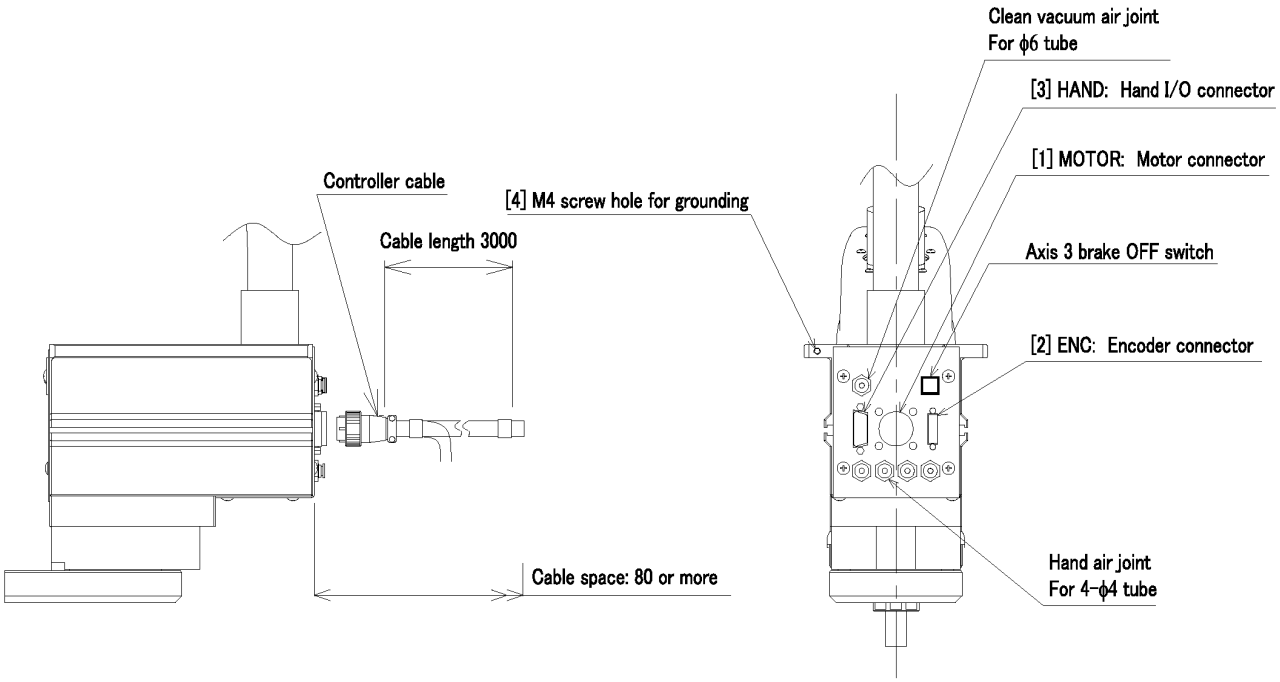


Fig. 3.4 Robot side connector arrangement (TH350A–T)

3.1.4 Connecting the Encoder Cable “ENC”

([3] of Fig. 3.1) (Cable attached)

The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The connector for connecting the encoder cable is ENC ([3] of Fig. 3.1). Location of the encoder cable on the robot side is [2] in Fig. 3.2, [2] in Fig. 3.3 or [2] in Fig. 3.4.

3.1.5 Connecting the Hand Control Signal Cable “HAND”

([4] of Fig. 3.1) (Cable: option)

The robot hand control signal cable is used for input and output of robot control signals such as hand operation signal.
The connector for connecting the hand control signal cable is HAND ([4] of Fig. 3.1). Location of the hand control signal cable on the robot side is [3] in Fig. 3.2, [3] in Fig. 3.3 or [3] in Fig. 3.4.

3.1.6 Grounding the Robot

An M4 screw hole for grounding purpose is provided on the basic robot base ([4] in Fig. 3.2 or [4] in Fig. 3.4).
Connect with the protective conductor of the entire facility completely.

3.1.7 Inserting and Removing Cables

! CAUTION

- Before inserting or removing any controller cable, be sure to turn off the "POWER" switch.
- When removing a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR

First, completely insert the connector on the cable side into the controller connector. Then, turn the cable side lock screw to the right to clamp the cable. A loose plug can cause a contact failure or other accident, so be careful and do the job right.

To remove the connector, turn the lock screw to the left and pull out the cable side connector.

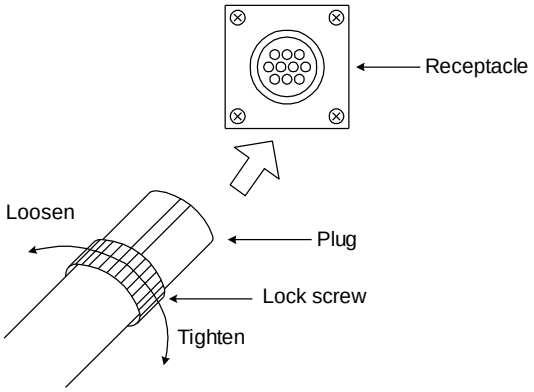


Fig. 3.5 Inserting and removing a circular connector

b) Square connectors: ENC, HAND, TP, INPUT, OUTPUT, COM1, HOST1, TCPRG

First, completely insert the cable side connector into the controller connector. Then, tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose plug can cause a contact failure or other accident, so be careful and do the job right.

To remove the connectors, first loosen the lock screws, then pull out the cable side connector.

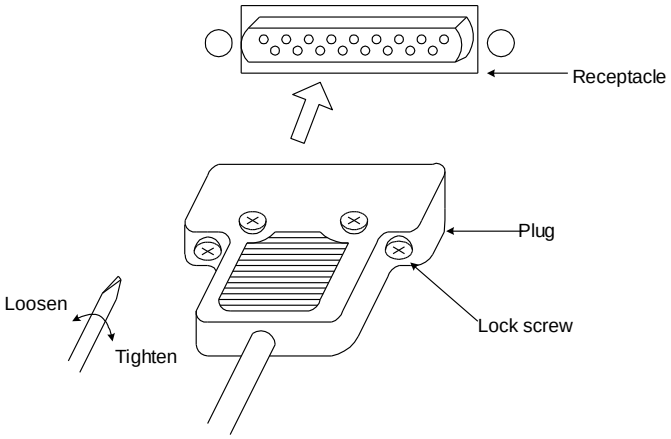
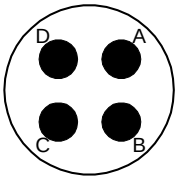
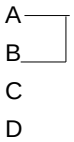


Fig. 3.6 Inserting and removing a square connector

3.1.8 Examples of Connector Terminal Arrangement

a) Power cable connector ACIN





Connects to controller.

Type: JL04V–2E18–10PE–B

Manufacturer:
Japan Aviation
Electronics Industry

Single phase, AC 170~250 V, 50/60 Hz

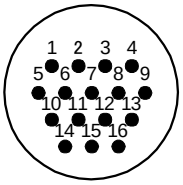
Grounding with grounding resistance of 100 Ω or less.

DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak has occurred. Or mis-operation may be caused by noise.

Fig. 3.7 External view of power cable connector

b) Motor cable connector MOTOR



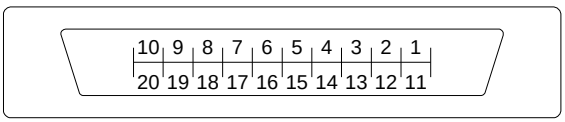
Connects to controller.

Type: JR25RK-16S

Manufacturer: Hirose Electric

Fig. 3.8 External view of motor cable connector

c) Encoder cable connector ENC



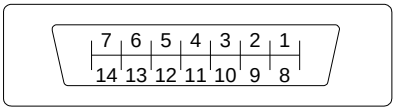
Connects to controller.

Type: 10120–3000VE

Manufacturer: 3M

Fig. 3.9 External view of encoder cable connector

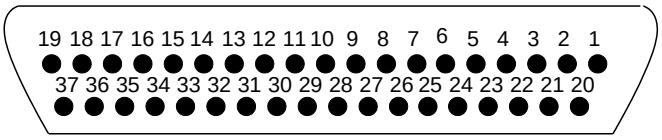
d) Robot hand control signal cable connector HAND



Connects to controller.
Type: 10114–3000VE
Manufacturer: 3M

Fig. 3.10 External view of robot hand control signal cable connector

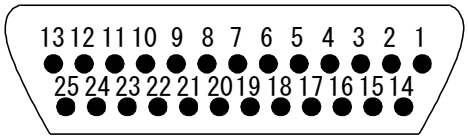
e) External input signal cable connector INPUT



Connects to controller.
Type: XM2F–3740–112L
Manufacturer: OMRON

Fig. 3.11 External view of external input signal cable connector

f) External output signal cable connector OUTPUT



Connects to controller.
Type: XM2F–3540–112L
Manufacturer: OMRON

Fig. 3.12 External view of external output signal cable connector

3.2 Controller Connector Signals

3.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	INPUT–12	(STOP)
	INPUT–14	(SV_OFF)
	INPUT–32	(BREAK)
	INPUT–18, 19	(EMS1B ~ EMS1C)
	INPUT–36, 37	(EMS2B ~ EMS2C)
	INPUT–17, 35	(INCOM3 ~ P24G)

These signals are already jumpered for the connectors provided for the TS1000 controller. If you wish to use them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system signals, be sure to connect the attached connector to the controller side INPUT connector.
It is recommended to jumper the following signals under some circumstances.

INPUT–13 (LOW_SPD)
INPUT–31 (CYCLE)

Table 3.2 Connector jumpers (P24G used as (–) common)

INPUT			
12–16	14–16	32–16	36–37
(13–16)	(31–16)	18–19	17–35

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- For all connectors not used, attach connector covers. Otherwise, the controller will malfunction due to static electricity.

Section 4

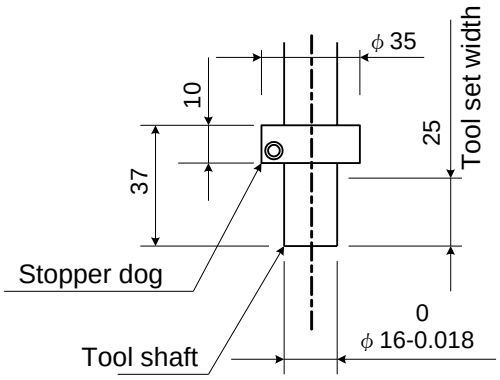
Tool Interface

4.1 Mounting Tool

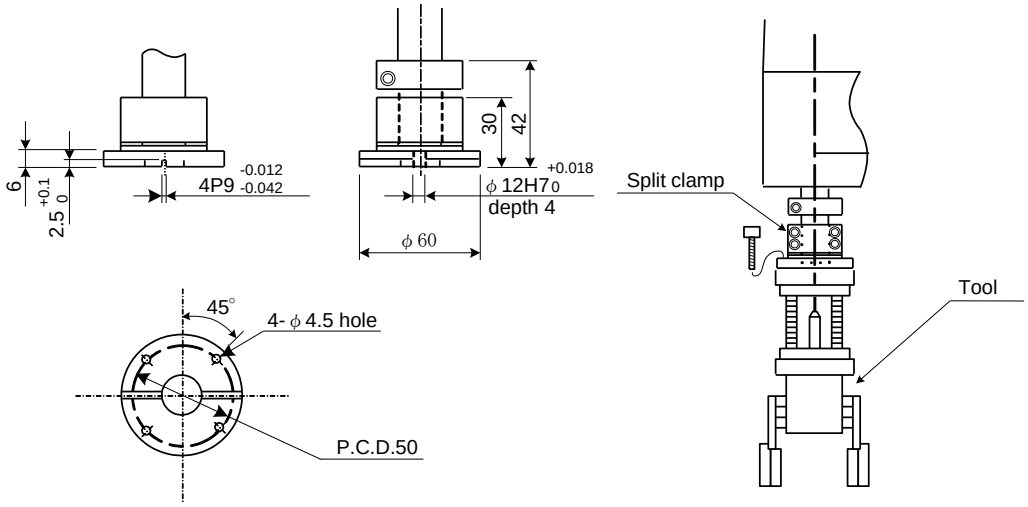
a) TH250A, TH350A, TH180 and TH350A‐T

The tool is mounted on the end of the tool shaft (ø16). The tool shaft has a dog for the stopper, and the section in front of the dog is used for mounting the tool. At this time, the dog should not be removed. Dimensions of the tool shaft section are shown in Fig. 4.1 (a).

As shown in Fig. 4.1, the tool is secured to the tool shaft with split clamp. As there is not a reference such as keyway on the tool shaft, however, take care of the tool direction when mounting.



(a) Dimensions of tool shaft section



(b) Dimensions of tool set flange (option)

Fig. 4.1 Mounting TH250A, TH350A , TH350A‐T and TH180 tools

4.2 Tool (Hand) Wiring and Piping

The robot is provided with wiring and air piping for the tool. These wiring and piping extend to the arm 2 from the base.

a) Wiring for tool

Arm side
DSUB 25-pin connector
(XM2D–1501 made by OMRON)

Pin No.	Signal name
1	LINE1
2	LINE2
3	LINE3
4	LINE4
5	LINE5
6	LINE6
7	LINE7
8	LINE8
9	LINE9
10	LINE10
11	LINE11
12	LINE12
13	LINE13
14	LINE14
15	Shield

Base side
DSUB 25-pin connector
(XM2A–1501 made by OMRON)

Pin No.	Signal name
1	LINE1
2	LINE2
3	LINE3
4	LINE4
5	LINE5
6	LINE6
7	LINE7
8	LINE8
9	LINE9
10	LINE10
11	LINE11
12	LINE12
13	LINE13
14	LINE14
15	Shield

Fig. 4.2 Machine I/O harness wiring

b) Air piping for tool

A total of four (4) lines ($\phi 4 \times 2.5$) are provided for the tool air piping, which are shown in Fig. 4.3 to Fig. 4.5.

Pneumatic devices such as solenoid valve should be provided by the customer.

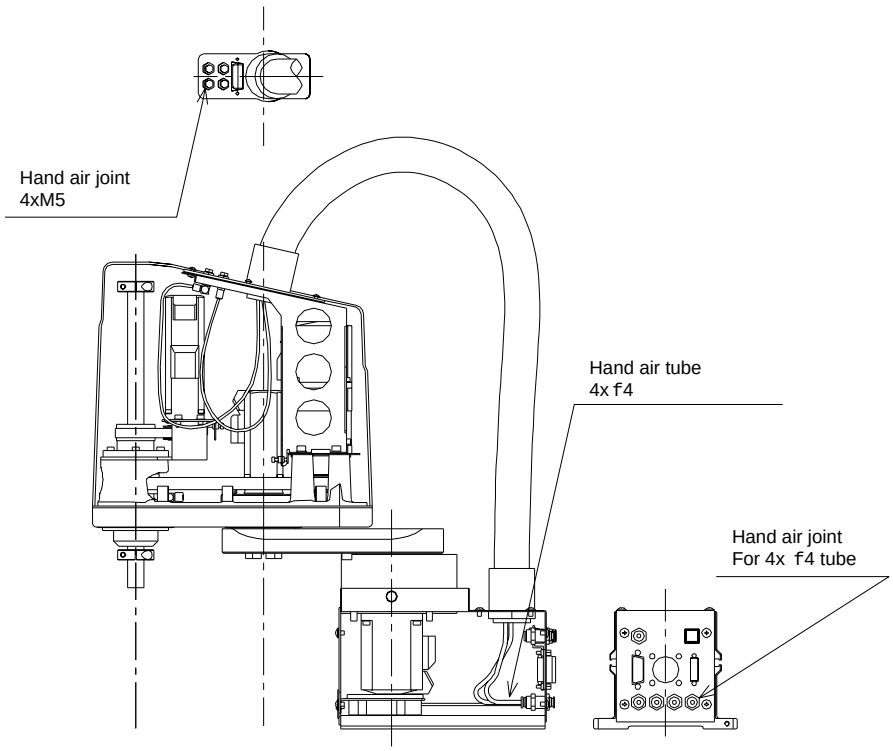


Fig. 4.3 Air piping for tool (TH250A, TH350A)

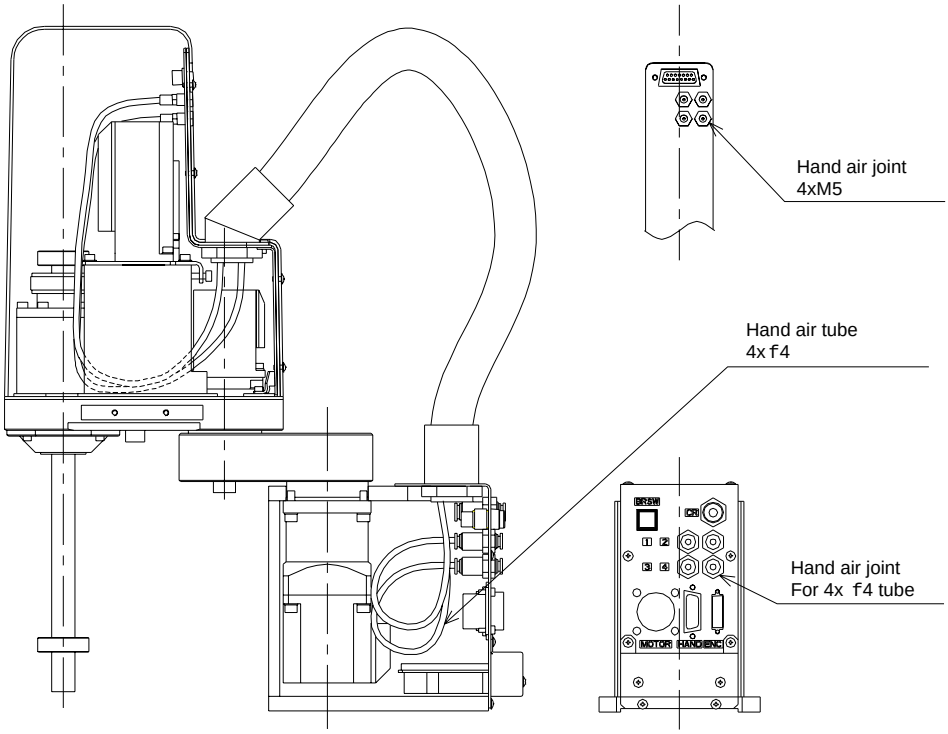


Fig. 4.4 Air piping for tool (TH180)

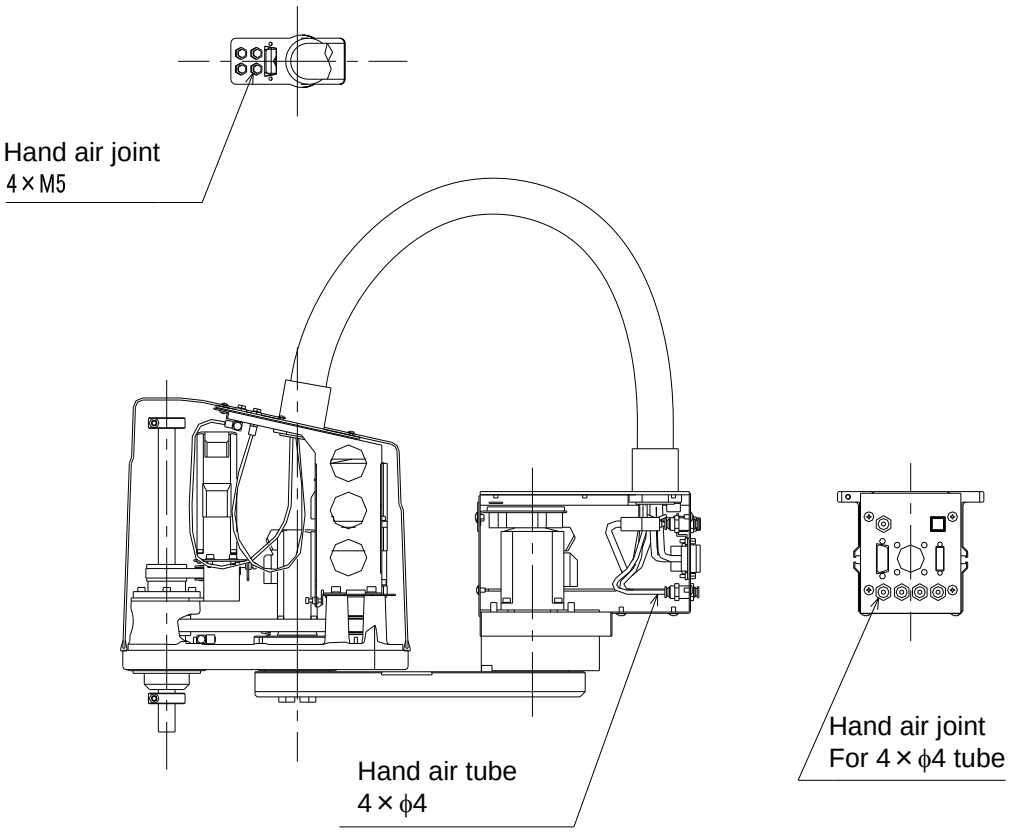


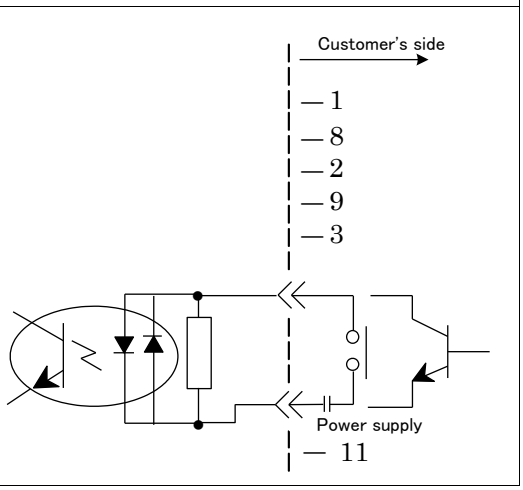
Fig. 4.5 Air piping for tool (TH350A－T)

4.2.1 Controller's Tool Signals

The controller is provided with tool signals (i.e., five (5) input signals for sensor, etc., four (4) control signals for solenoid valve, etc., DC 24V and DC 24 VGND signals), which can be connected also from the controller side. (A cable between the controller and cable is an option.) The relevant signals are described below.

a) Input signal connector HAND

Table 4.1 Input signals

Pin	Signal name		Signal No.	Input circuit and example of connections
1	HI_2	Hand input 1	201	
8	HI_2	Hand input 2	202	
2	HI_3	Hand input 3	203	
9	HI_4	Hand input 4	204	
3	HI_5	Hand input 5	205	
10	P24V	24 V		
4	P24G	0 V		
11	HI_COM1	Common		

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: DC24 V, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

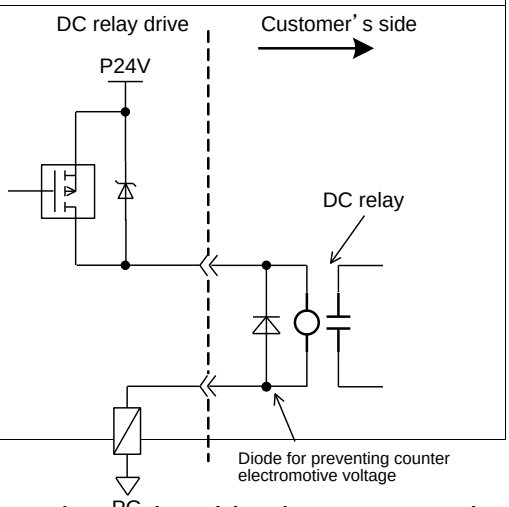
Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

b) Output signal connector HAND

Table 4.2 Output signals

Pin	Signal name		Signal No.	Output circuit and example of connections
5	HO_1	Hand output 1	201	
12	HO_2	Hand output 2	202	
6	HO_3	Hand output 3	203	
13	HO_4	Hand output 4	204	
7	P24V	DC24 V power GND		
14	P24G	DC24 V power GND		

By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the TS1000 robot controller. NEVER connect +24 V with P24 V. Otherwise, the power supply may be damaged.

Output specifications:

Rated voltage : DC24 V (max. DC30 V)
Rated current : 1 A (1.2 A or less including other I/Os when the internal power supply is used.)
Leak current : 100 µA or less

- If the DC24 V power supplied from the robot controller is used, the total current should be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.
- When a double solenoid is used, HO_1 and HO_2, and HO_3 and HO_4 are used as pairs.

The figure below shows the DC relay circuit when the external power is used.

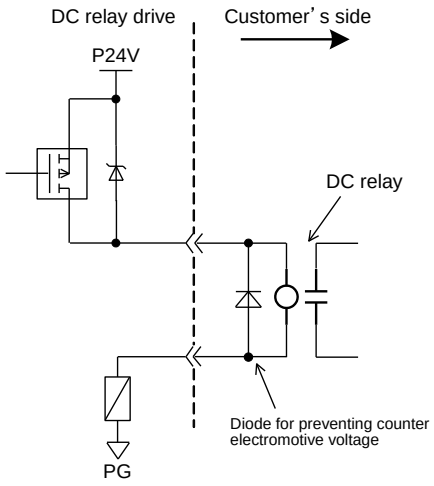


Fig. 4.6 DC relay circuit diagram

4.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

4.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 4.7and Fig. 4.8.
The load conditions permitted for the TH250A, TH350A, TH180 and TH350A‐T robots are shown in Table 4.3.

Table 4.3 Permissible load conditions

Model	Conditions	Permissible values
TH250A TH350A TH350A‐T	Load mass	Max. 3 kg
	Moment of inertia	Max. 0.0170 kg·m^2
	Gravity center offset of load	Max. 70 mm
TH180	Load mass	Max. 2 kg
	Moment of inertia	Max. 0.01 kg·m^2
	Gravity center offset of load	Max. 50 mm

! CAUTION

- The robot life and safety cannot be guaranteed if the robot is operated under the load conditions exceeding the permissible values.

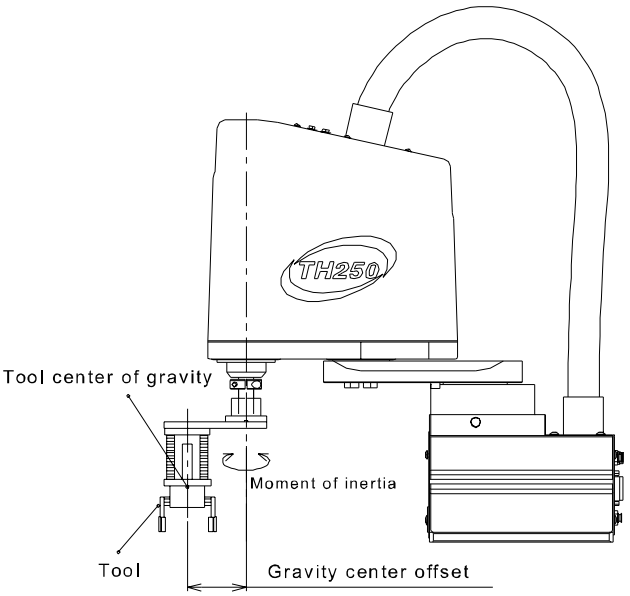


Fig. 4.7 Tool and robot body (Normal type)

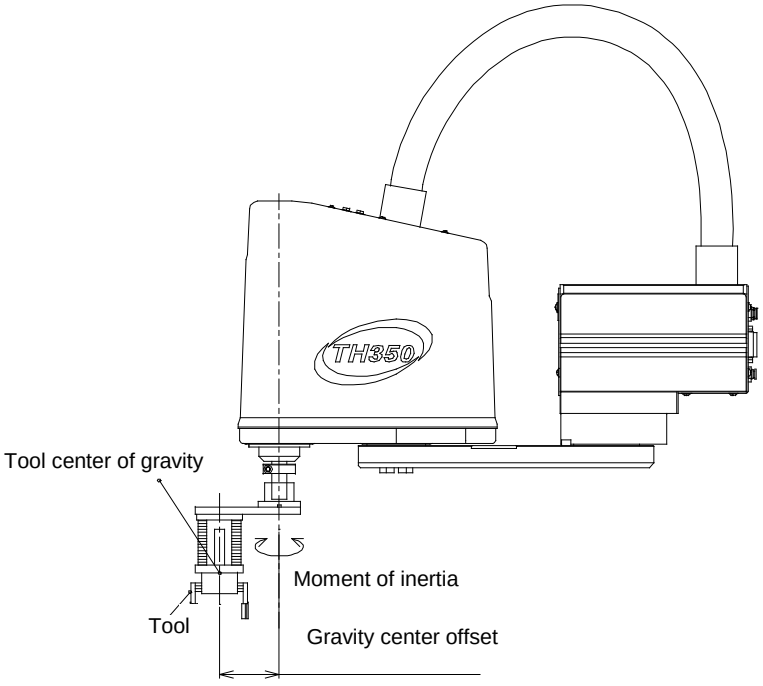


Fig. 4.8 Tool and robot body (Ceiling type)

4.3.2 Load Conditions and Program Setting

Even if the tool load conditions are within the permissible values listed in Table 4.3, acceleration and servo gain that are adequate for the load conditions should be set. If this setting is inadequate, the robot may vibrate or overshoot, resulting in reduction of the robot life.
The TH series can automatically change the acceleration and servo gain according to the load conditions by using the PAYLOAD command in the program. The specific method for using this function is explained below.

a) PAYLOAD command
The PAYLOAD command format is written as shown below if the tool mass is M kg and the gravity center is L mm.

PAYLOAD = {M, L}

M : Load mass (unit: kg)
L : Gravity center offset (unit: mm)

The PAYLOAD command has the following functions.

- The acceleration of each robot axis is automatically changed according to the set load conditions.
- The servo gain of each robot axis is automatically changed according to the set load conditions.

b) Example programs
The basic programs using the PAYLOAD command are exemplified below. For further information, see the Robot Language Manual.

(Example program 1)
The robot is moved under the load conditions of 2 kg mass and 30 mm gravity center offset.

```
PROGRAM SAMPLE
  SPEED= 100
  PAYLOAD={2,30}
  MOVE A
  MOVE B
  STOP
END
```

(Program example 2)

Let’s assume that the hand mass is 1 kg and gravity center offset is 30 mm, and that the mass is 3 kg and gravity center offset is 50 mm when the workpiece is grasped.

Pick-and-place operation is executed under these conditions.

```
PROGRAM SAMPLE
  PAYLOAD={1,30}
  ACCUR=COARSE
  ENABLE NOWAIT
  RESET DOUT
  MOVE TAIKI
  DOUT(1)
  WAIT DIN(1)
LOOP:
  MOVE A+POINT(0,0,100)
  IF DIN(–1)THEN GOTO FIN
  MOVE A
  WAIT MOTION>=100
  DOUT(213)
  DELAY 1
  PAYLOAD={3,50}
  MOVE A+POINT(0,0,100)
  MOVE B+POINT(0,0,100)
  MOVE B
  WAIT MOTION>=100
  DOUT(–213)
  DELAY 1
  PAYLOAD={1,30}
  MOVE B+POINT(0,0,100)
  GOTO LOOP
FIN:
  MOVE TAIKI
  DOUT(1)
  STOP
END
```

c) Setting of PAYLOAD command

In the default state, or when the PAYLOAD command is not used, the acceleration is set to 100 % and the servo gain is set to the value under the minimum load. (See Para. 4.3.3.)

! CAUTION

Be sure to use the PAYLOAD command in the following occasions.

- When the load mass exceeds 1 kg.
- When there is a gravity center offset of load.

Unless the PAYLOAD command is used, the robot will vibrate or overshoot, resulting in shortening of the life of the mechanisms.

! CAUTION

The moment of inertia of load should be within the tolerances given in Table 4.3. Even if there is no gravity center offset of load, the robot may vibrate if the moment of inertia is large. When this happens, figure out virtual gravity center offset L [mm] from the following equation, using the moment of inertia J [kg·m^2] and mass M [kg].

$L = \sqrt{(J \times 10^6 / M)}$

Then, designate the following.

PAYLOAD = {M, L}

! CAUTION

When guiding manually, the robot may vibrate if the load mass or gravity center offset is large. This is because the servo gain is not appropriate. When this happens, perform the following operation while setting the load conditions in the TEACHING mode.

DO PAYLOAD = {5, 100}
EXE

Then, the servo gain is changed to the value which meets the load conditions.

If the robot still vibrates or overshoots after setting the above conditions, contact SHIBAURA MACHINE. Advise us of the then load conditions and program also.

4.3.3 Setting Robot Acceleration and Maximum Speed for Load Conditions

a) TH250A

Acceleration of the robot is automatically changed according to the load conditions when the PAYLOAD command is used.

The acceleration changes with the load mass, as shown in Fig. 4.9. The vertical line shows the acceleration. If the load mass is 3 kg, for instance, the acceleration of axes 1, 2, and 4 is 66 % of the maximum value.

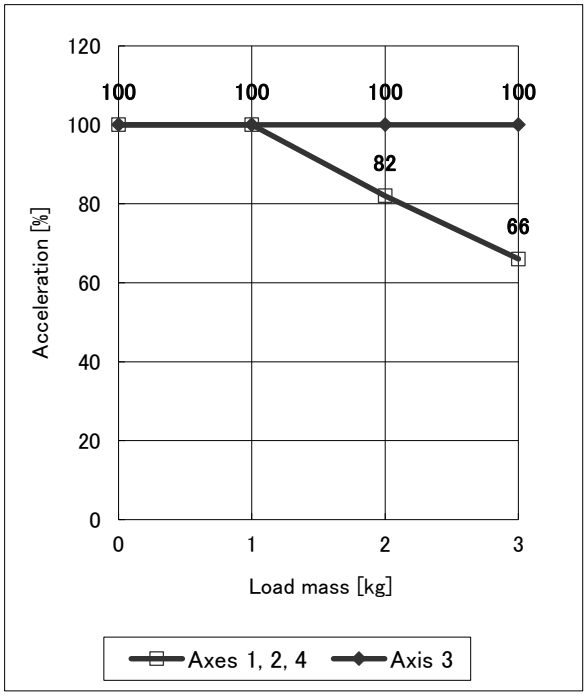


Fig. 4.9 Setting of acceleration for load mass (TH250A)

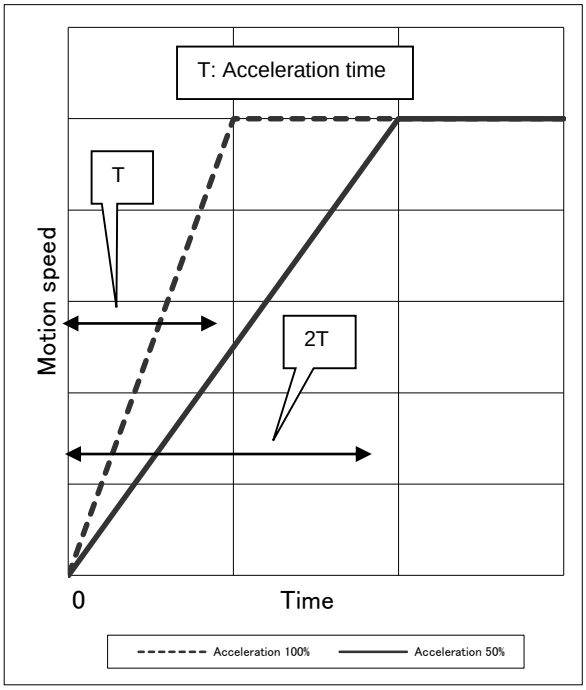
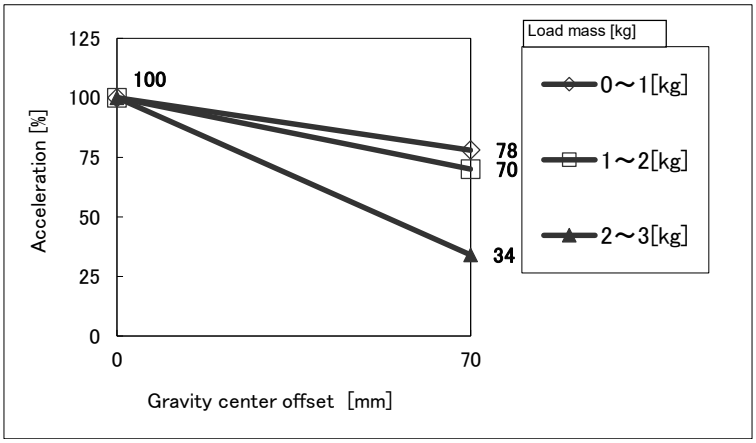
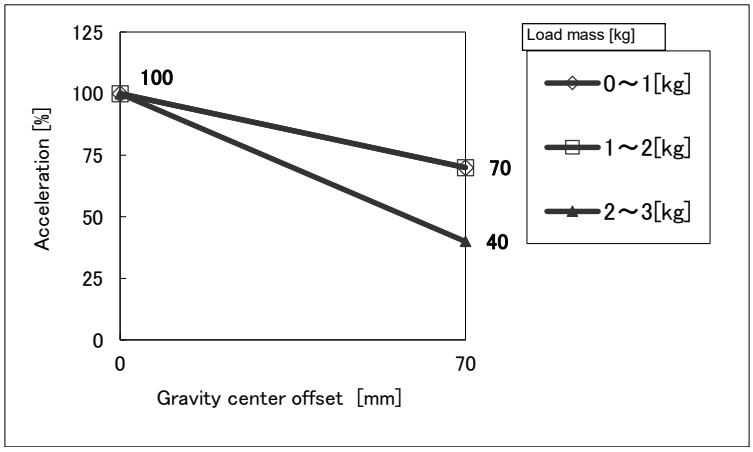


Fig. 4.10 Acceleration changes

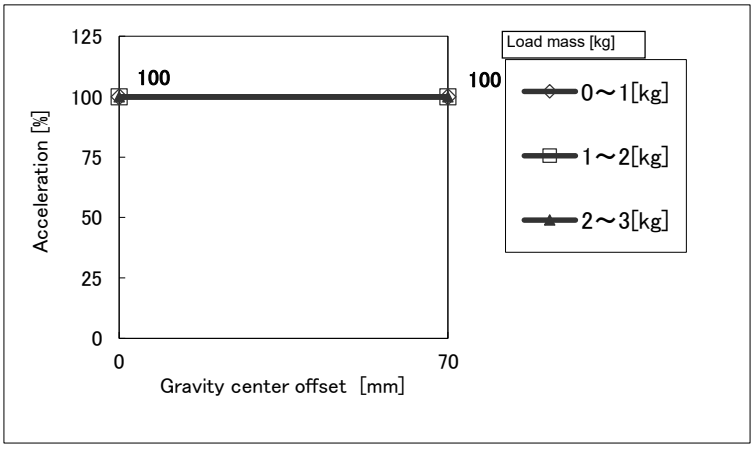
Additionally, if there is a gravity center offset of load, the acceleration and maximum speed change as shown in Fig. 4.11 and Fig. 4.12, respectively. If the load mass is 3 kg and gravity center offset is 70 mm, for instance, the acceleration of axes 1 and 2 is 22 % (= 66 % × 34 %) and the maximum speed of axes 1, 2 and 4 is 100 %.



(a) Axes 1 and 2

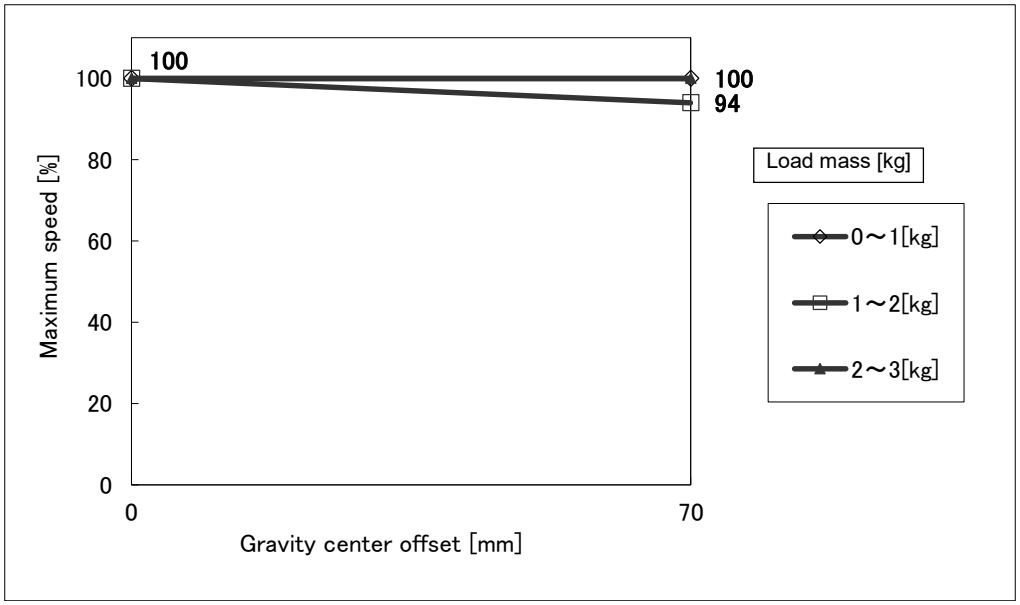


(b) Axis 4

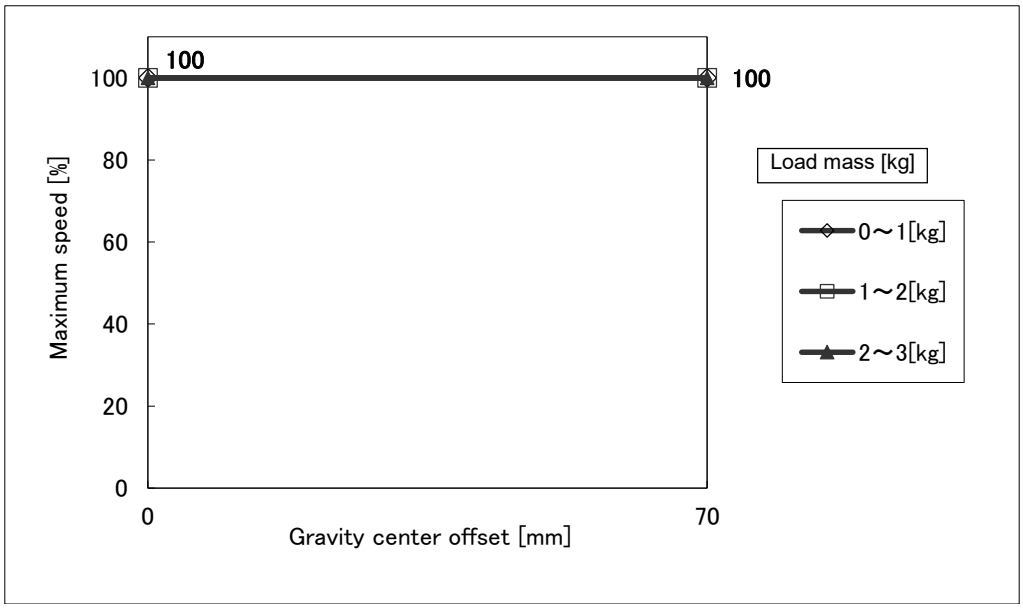


(c) Axis 3

Fig. 4.11 Setting of acceleration for gravity center offset (TH250A)



(a) Axes 1, 2, 4



(b) Axis 3

Fig. 4.12 Setting of maximum speed for gravity center offset (TH250A)

b) TH350A

Acceleration of the robot is automatically changed according to the load conditions when the PAYLOAD command is used.

The acceleration changes with the load mass, as shown in Fig. 4.13. The vertical line shows the acceleration. If the load mass is 3 kg, for instance, the acceleration of axes 1, 2 and 4 is 70 %.

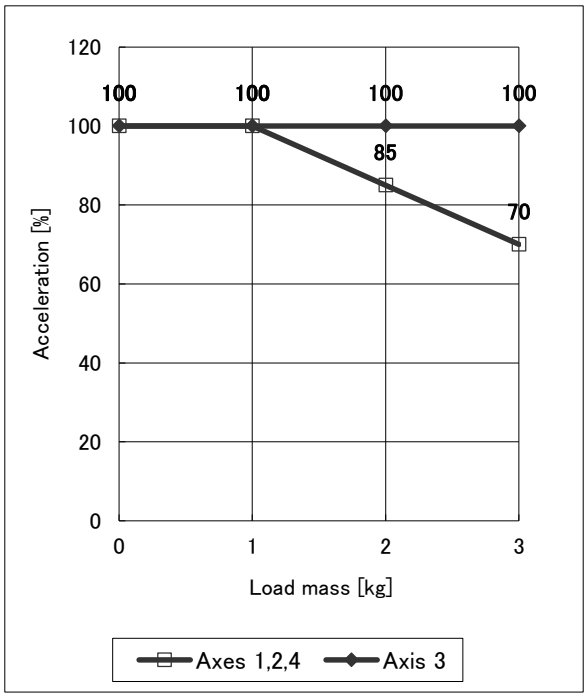
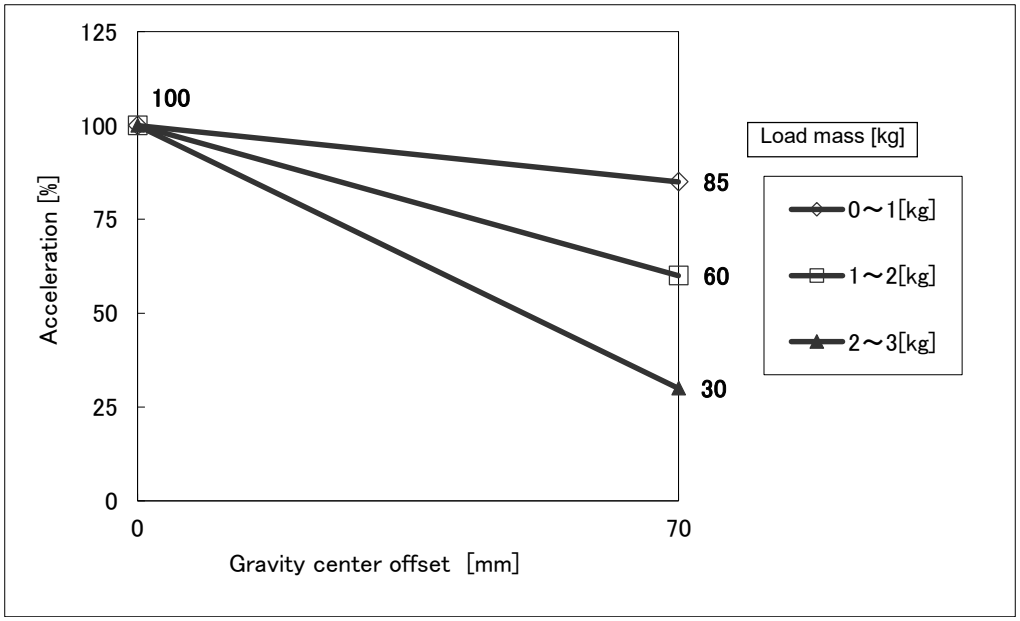


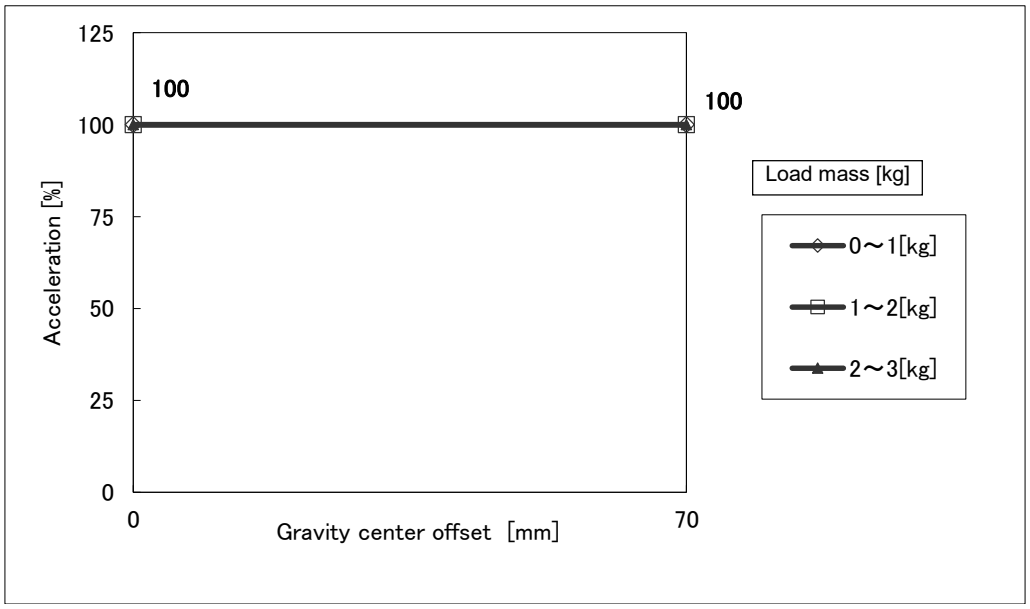
Fig. 4.13 Setting of acceleration for load mass (TH350A)

Additionally, if there is a gravity center offset of load, the acceleration changes as shown in Fig. 4.14.

If the load mass is 3 kg and gravity center offset is 70 mm, for instance, the acceleration of axes 1, 2 and 4 is 21 % (= 70 % × 30 %).



(a) Axes 1, 2, and 4



(b) Axis 3

Fig. 4.14 Setting of acceleration for gravity center offset (TH350A)

c) TH180

Acceleration of the robot is automatically changed according to the load conditions when the PAYLOAD command is used.

The acceleration changes with the load mass, as shown in Fig. 4.15. The vertical line shows the acceleration. If the load mass is 2 kg, for instance, the acceleration of axes 1 and 2 is 79 %.

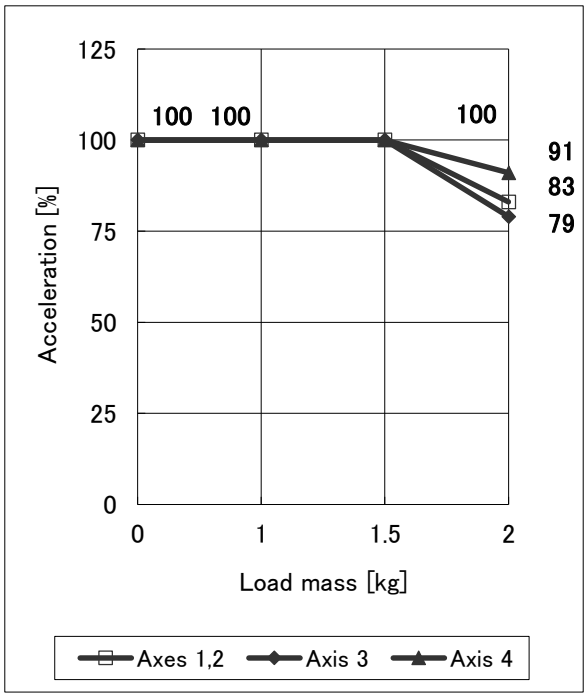
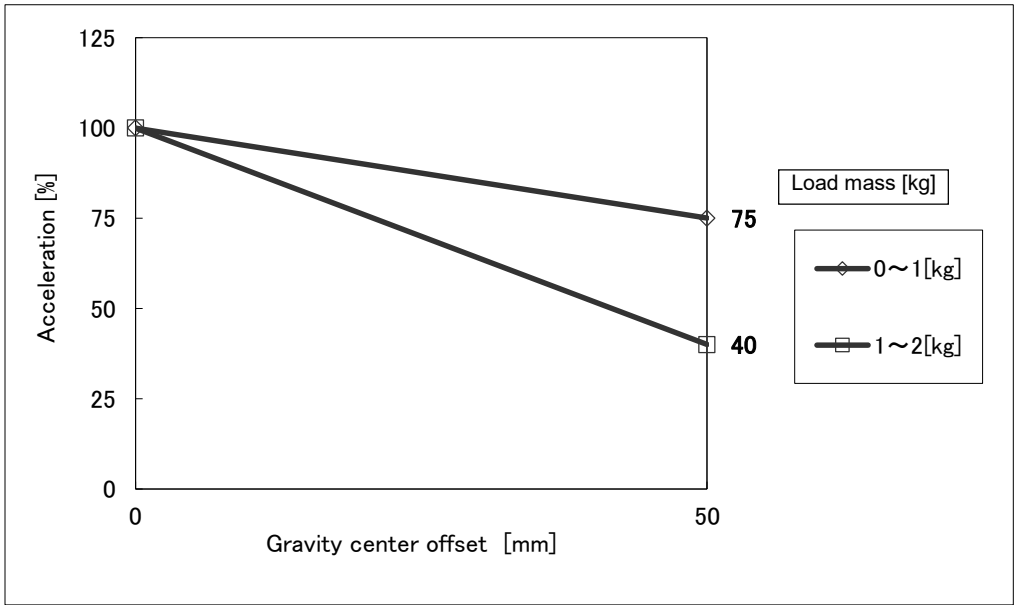


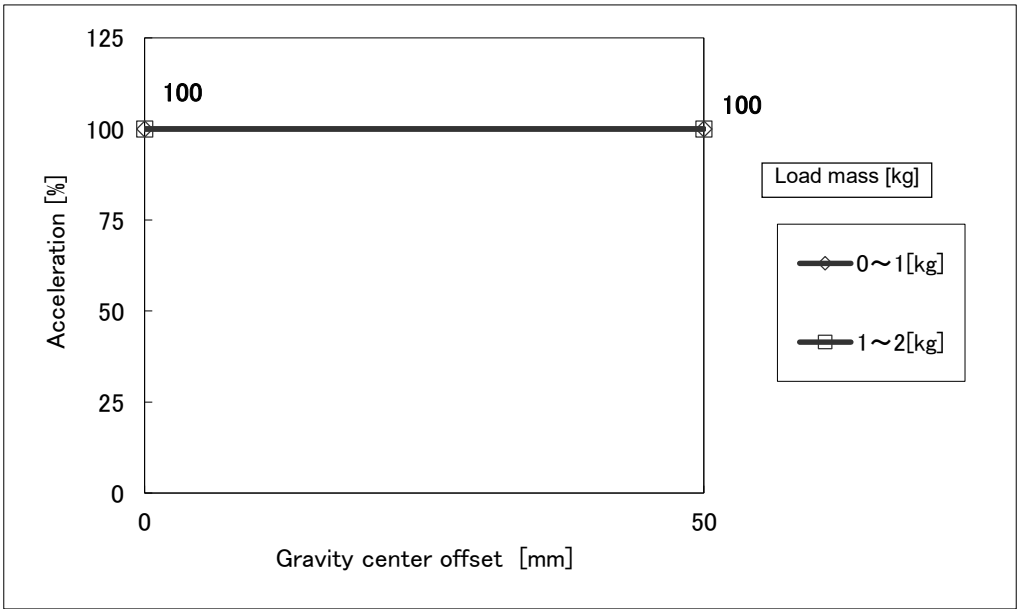
Fig. 4.15 Setting of acceleration for load mass (TH180)

Additionally, if there is a gravity center offset of load, the acceleration changes as shown in Fig. 4.16.

If the load mass is 2 kg and gravity center offset is 50 mm, for instance, the acceleration is 31 % (= 79 % × 40 %).



(a) Axes 1, 2, and 4



(b) Axis 3

Fig. 4.16 Setting of acceleration for gravity center offset (TH180)

d) TH350A－T

Acceleration of the robot is automatically changed according to the load conditions when the PAYLOAD command is used.

The acceleration changes with the load mass, as shown in Fig. 4.17. The vertical line shows the acceleration. If the load mass is 3 kg, for instance, the acceleration of axes 1, 2 and 4 is 70 %.

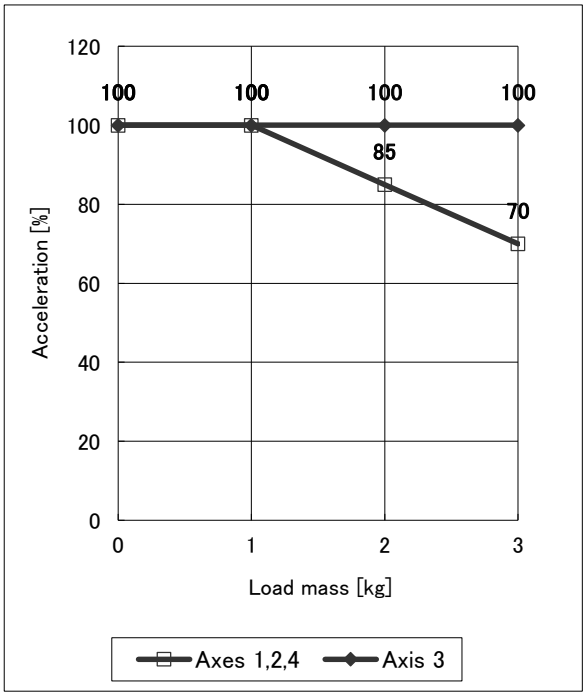
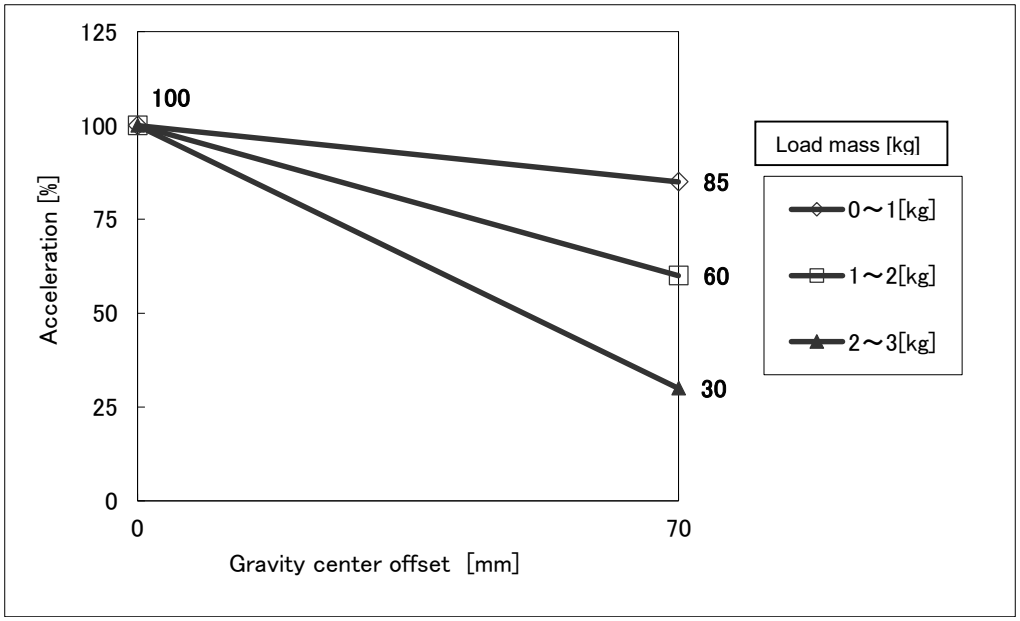


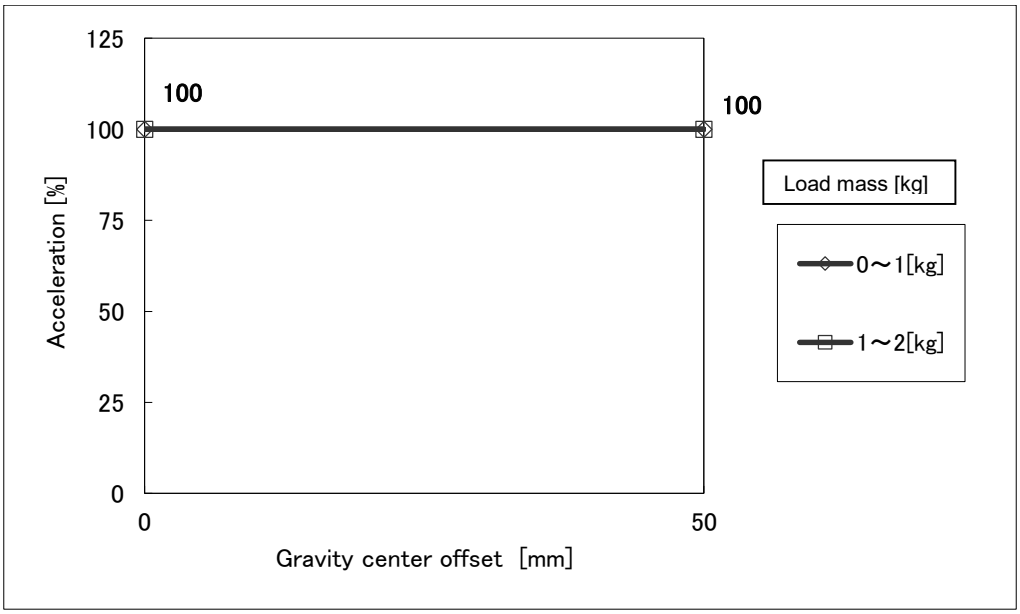
Fig. 4.17 Setting of acceleration for load mass (TH350A－T)

Additionally, if there is a gravity center offset of load, the acceleration changes as shown in Fig. 4.18.

If the load mass is 3 kg and gravity center offset is 70 mm, for instance, the acceleration of axes 1, 2 and 4 is 21 % (= 70 % × 30 %).



(a) Axes 1, 2, and 4



(b) Axis 3

Fig. 4.18 Setting of acceleration for gravity center offset (TH350A‐T)

Section 5

Specifications

5.1 Specifications Table

Item		Specifications
Structure		Horizontal multi-joint type SCARA robot
Model		TH180
Applicable controller		TS1000
Mass of actuator		9 kg
No. of controlled axes		Four (4)
Arm length		180 mm (70 mm + 110 mm)
Motor capacity	Axis 1	100 (W)
	Axis 2	50 (W)
	Axis 3	50 (W)
	Axis 4	50 (W)
Operating range	Axis 1	±120 (deg)
	Axis 2	±140 (deg)
	Axis 3	120 (mm)
	Axis 4	±360 (deg)
Maximum speed (*1)	Axis 1	533 (deg/s)
	Axis 2	480 (deg/s)
	Axis 3	1013 (mm/s)
	Axis 4	1186 (deg/s)
	Composite speed of axes 1 and 2	2.6 (m/s)
Rated payload mass		1 (kg)
Maximum payload mass		2 (kg)
Permissible load inertia (*1)		0.01 ($kg \cdot m^2$)
Repeatability	X, Y	±0.01 (mm)
	Z	±0.01 (mm)
	C	±0.005 (deg)
Cycle time (*2) (When payload mass is 2 kg)		0.35 (sec)
Drive system		By means of AC servo motors
Position detection method		Absolute

*1: When the mass of load exceeds 1 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Item		Specifications	
Structure		Horizontal multi-joint type SCARA robot	
Model		TH250A	TH350A
Applicable controller		TS1000	
Mass of actuator		14 kg	
No. of controlled axes		Four (4)	
Arm length		250 mm (125 mm + 125 mm)	350 mm (225 mm + 125 mm)
Motor capacity	Axis 1	200 (W)	
	Axis 2	100 (W)	
	Axis 3	100 (W)	
	Axis 4	100 (W)	
Operating range	Axis 1	±115 (deg)	
	Axis 2	±140 (deg)	
	Axis 3	120 (mm)	
	Axis 4	±360 (deg)	
Maximum speed (*1)	Axis 1	540 (deg/s)	337.5 (deg/s)
	Axis 2	500 (deg/s)	
	Axis 3	1120 (mm/s)	
	Axis 4	1143 (deg/s)	
	Composite speed of axes 1 and 2	3.53 (m/s)	3.24 (m/s)
Rated payload mass		1 (kg)	
Maximum payload mass		3 (kg)	
Permissible load inertia (*1)		0.017 (kg·m^2)	
Repeatability	X, Y	±0.01 (mm)	
	Z	±0.01 (mm)	
	C	±0.03 (deg)	
Cycle time (*2) (When payload mass is 2 kg)		0.41 (sec)	
Drive system		By means of AC servo motors	
Position detection method		Absolute	

*1: When the mass of load exceeds 1 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Appendix A Calculating Control Power Capacity

The power capacity of the controller can be calculated in the following manner. The power connected to the controller should exceed the capacity obtained from the following equation.

(Power capacity) = (Power capacity of control unit) + (Power capacity of servo driver)

a) Power capacity of control unit
The power capacity of the controller excluding the servo driver is as follows:
Power capacity of control unit ..... 208 VA

b) Power capacity of servo driver
The power capacity of the servo driver is the total of the power capacities of the servo motors used in the robot. The following table lists the power capacity of each motor.

Table 5.1 Motor power capacity

Motor capacity (W)	Power capacity (VA)
50	71
100	143
200	286

Example: The servo motor capacity and servo driver power capacity of the TH series robots are as follows:

Table 5.2 Servo driver power capacity

Type of robot	Axis 1	Axis 2	Axis 3	Axis 4	Power capacity of servo driver (VA)
TH250A	200	100	100	100	286 + 143 + 143 + 143 = 715
TH350A	200	100	100	100	286 + 143 + 143 + 143 = 715
TH180	100	50	50	50	143 + 71 + 71 + 71 = 356
TH350A‐T	200	100	100	100	286 + 143 + 143 + 143 = 715

Reference: The power capacity of each servo motor is calculated by the following equation.

(Power capacity of servo driver) = Servo rated output / 0.7

c) Example of calculating power capacity

The power capacity of TH series robot is listed in the following table.

Table 5.3 Power capacity of TH series robots

Type of robot	Control unit power capacity [VA]	Servo driver power capacity [VA]	Controller power capacity [VA]
TH250A	208	715	923
TH350A	208	715	923
TH180	208	356	564
TH350A‐T	208	715	923

Note: The above power capacity is required when all axes are continuously operated under 100% load. If the power capacity is insufficient when the robot is operated, troubles such as a drop in master power voltage, a malfunction, or resetting of the control power may take place. Thus, the robot may not be correctly operated.

Also, an electric current that is three (3) times the rated current may be required at the start of the robot. Thus, it is necessary to select a power supply in which the peak current of the power is three (3) times the rated current for 100 ms.

APPROVED BY: Y. Yamaguchi

CHECKED BY: H. SAITO

PREPARED BY: H. Yamano